Health Information System

Prepared by

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Acknowledgments

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Contents

Course Description

This course is intended to provide students with a broad understanding of the concepts of information systems, technical aspects of information systems and applications of information systems in healthcare. It provides an overview of the healthcare information systems industry, its history, recent developments and continuing challenges, as well as a practical understanding of healthcare information systems acquisition and implementation.

Core Knowledge

By the end of this course, students should be able to:

- Understanding of the concepts of information systems
- Diagram the information systems lifecycle and its process and critical issues
- Define the health information systems (HIS)
- Know the history of HIS
- State the various types of HIS
- Determine the challenges of HIS
- Describe the data quality parameter

Core Skills

By the end of this course, students should be able to:

- Apply IT tools and approaches to health care field
- Discover what makes successful HIS.
- Compare the different types of HIS
- Report the important of HIS and its rules in human will being
- Modify the methods used in communicating and exchanging health information
- Analyze obstacles and success factors for implementation and integration of information, communication and decision technologies in healthcare
- Show the technical and policy implications of HIS

- Relate the use of data and enabling information technologies healthcare field
- Evaluate the roles of information system applications common in healthcare organization
- Conduct an information technology needs assessment of a healthcare organizational unit.
- Show good knowledge of application software; including spreadsheets; e-mail; word processing; database management.

Course Overview

		Methods of Teaching / Training with Number of Total Hours per Topic				
ID	**Topics**	**Interactive Lecture**	**Field Work**	**Class Assignments**	**Research**	**Lab**
1	Introduction to Information system	2		2		
2	Introduction to Health Information system	2		2		
3	Health care data quality	2		2		
4	History and evaluation of Health Information Systems	2		2		
5	Architecture of a Health Information System	2		2		
6	Challenges for Health Information Systems	2		2		
7	Types of Health Information Systems	2		2		
8	Technology that support health information systems	2		2		
9	Electronic Health Records as a Part of Health Information Systems	2		2		
10	Data privacy and Security of health information systems	2		2		
11	e-governance and management	2		2		
12	Applications of Health Information Systems	2		2		
TOTAL HOURS (48)		**24**		**24**		

Chapter 1

Introduction to Information Systems

Objectives

- Understanding of the concepts of information systems
- Diagram the information systems lifecycle and its process and critical issues
- Discuss why it is important to study and understand information systems.
- Distinguish data from information and describe the characteristics used to evaluate the value of data.
- Name the components of an information system and describe several system characteristics.
- List the components of a computer-based information system.

WHAT IS AN INFORMATION SYSTEM?

An information system (IS) is a set of interrelated elements or components that collect (input), manipulate (process), ***store, and disseminate (output) data and information, and provide a corrective reaction (feedback mechanism) to*** meet an objective (see Figure 1). The feedback mechanism is the component that helps organizations achieve their goals, such as increasing profits or improving customer service.

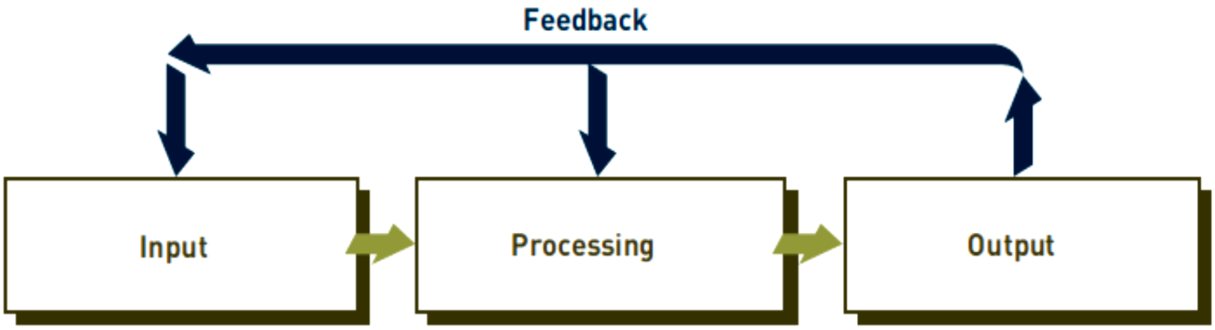


Figure 1: Information system component

- **Input:** In information systems, **input** is the activity of gathering and capturing raw data. In producing paychecks, for example, the number of hours every employee works must be collected before paychecks can be calculated or printed.

- **Processing:** In information systems, **processing** means converting or transforming data into useful outputs. Processing can involve making calculations, comparing data and taking alternative actions, and storing data for future use. Processing data into useful information is critical in business settings. Processing can be done manually or with computer assistance. In a payroll application, the number of hours each employee worked must be converted into net, or take-home, pay. Other inputs often include employee ID number and department. The processing can first involve multiplying the number of hours worked by the employee's hourly pay rate to get gross pay. If weekly hours worked exceed 40, overtime pay might also be included. Then deductions—for example, federal and state taxes, contributions to insurance or savings plans—are subtracted from gross pay to get net pay. After these calculations and comparisons are performed, the results are typically stored. *Storage* involves keeping data and information available for future use, including output, discussed next.
- **Output:** In information systems, **output** involves producing useful information, usually in the form of documents and reports. Outputs can include paychecks for employees, reports for managers, and information supplied to stockholders, banks, government agencies, and other groups. In some cases, output from one system can become input for another. For example, output from a system that processes sales orders can be used as input to a customer billing system.
- **Feedback:** In information systems, **feedback** is information from the system that is used to make changes to input or processing activities. For example, errors or problems might make it necessary to correct input data or change a process. Consider a payroll example. Perhaps the number of hours an employee worked was entered as 400 instead of 40. Fortunately, most information systems check to make sure that data falls within certain ranges. Feedback is also important for managers and decision makers. For example, a furniture maker could use a computerized feedback system to link its suppliers and plants. The output from an information system might indicate that inventory levels for mahogany and oak are getting low—a potential problem.

Manual and Computerized Information Systems

An information system can be manual or computerized. For example, some investment analysts manually draw charts and trend lines to assist them in making investment decisions. Tracking data on stock prices (input) over the last few months or years, these analysts develop patterns on graph paper (processing) that help them determine what stock prices are likely to do in the next few days or weeks (output).

Computer-Based Information Systems

A computer-based information system (CBIS) is a single set of hardware, software, databases, telecommunications, people, and procedures that are configured to collect, manipulate, store, and process data into information. Some new cars and home appliances include computer hardware, software, databases, and even telecommunications to control their operations and make them more useful. This is often called embedded, pervasive, or ubiquitous computing.

The components of a CBIS are illustrated in Figure 2, Information technology (IT) refers to hardware, software, databases, and telecommunications. A business's technology infrastructure includes all the hardware, software, databases, telecommunications, people, and procedures that are configured to collect, manipulate, store, and process data into information. The technology infrastructure is a set of shared IS resources that form the foundation of each computer-based information system.

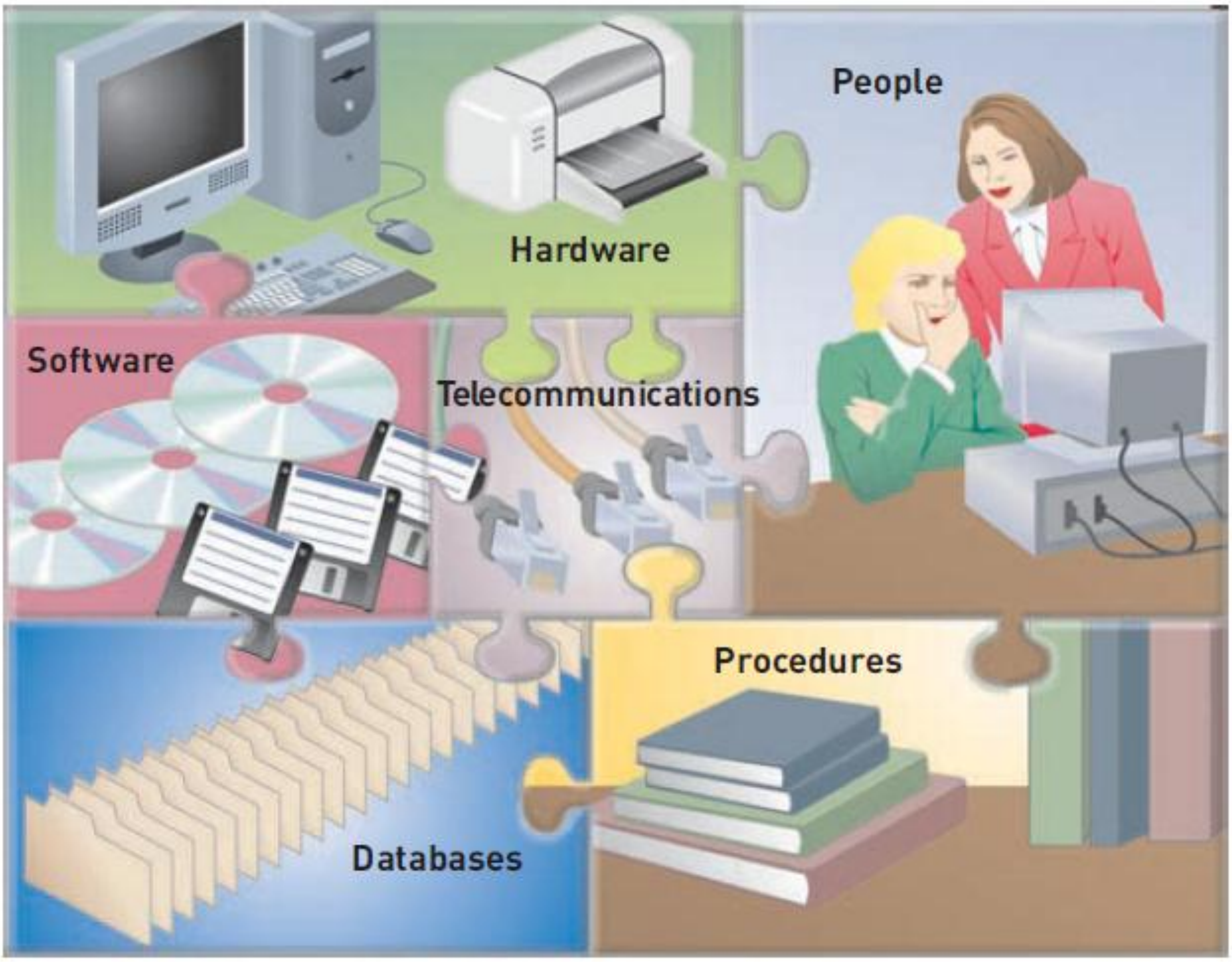


Figure 2: The components of a CBIS

- **Hardware** consists of computer equipment used to perform input, processing, and output activities.

- **Software** consists of the computer programs that govern the operation of the computer. These programs allow a computer to process payroll, send bills to customers, and provide managers with information to increase profits, reduce costs, and provide better customer service. With software, people can work anytime at any place.

- **Database** is an organized collection of facts and information, typically consisting of two or more related data files. An organization's database can contain facts and information on customers, employees, inventory, competitors' sales, online purchases, and much more.

- **Telecommunications** is the electronic transmission of signals for communications, which enables organizations to carry out their processes and tasks through effective computer networks.

- **Networks** connect computers and equipment in a building, around the country, or around the world to enable electronic communication. Investment firms can use wireless networks to connect thousands of investors with brokers or traders. Many hotels use wireless telecommunications to allow guests to connect to the Internet, retrieve voice messages, and exchange e-mail without plugging their computers or mobile devices into an Internet connector. **Internet** is the world's largest computer network, consisting of thousands of interconnected networks, all freely exchanging information. Research firms, colleges, universities, high schools, and businesses are just a few examples of organizations using the Internet. People use the Internet wherever they are to research information,

- **People** can be the most important element in most computer-based information systems. They make the difference between success and failure for most organizations. Information systems personnel include all the people who manage, run, program, and maintain the system. Large banks can hire IS personnel to speed the development of computer-related projects. Users are people who work with information systems to get results. Users include financial executives, marketing representatives, manufacturing operators, and many others. Certain computer users are also IS personnel.

- **Procedures** include the strategies, policies, methods, and rules for using the CBIS, including the operation, maintenance, and security of the computer. For example, some procedures describe when each program should be run. Others describe who can access facts in the database or what to do if a disaster, such as a fire, earthquake, or hurricane, renders the CBIS unusable. Good procedures can help companies take advantage of new opportunities and avoid potential disasters. Poorly developed and inadequately implemented procedures, however, can cause people to waste their time on useless rules or result in inadequate responses to disasters, such as hurricanes or tornadoes.

SYSTEMS DEVELOPMENT

Systems development is the activity of creating or modifying business systems. Systems development projects can range from small to very large and are conducted in fields as diverse as stock analysis and video game development. Some systems development efforts are a huge success.

People inside a company can develop systems, or companies can use outsourcing, hiring an outside company to perform some or all of a systems development project. Outsourcing allows a company to focus on what it does best and delegate other functions to companies with expertise in systems development. Outsourcing, however, is not the best alternative for all companies. Developing information systems to meet business needs is highly complex and difficult—so much so that it is common for IS projects to overrun budgets and exceed scheduled completion dates. One strategy for improving the results of a systems development project is to divide it into several steps, each with a well-defined goal and set of tasks to accomplish (see Figure 3). These steps are summarized next.

Systems Investigation and Analysis

The first two steps of systems development are systems investigation and analysis. The goal of the systems investigation is to gain a clear understanding of the problem to be solved or opportunity to be addressed. After an organization understands the problem, the next question is, "Is the problem worth solving?" Given that organizations have limited resources—people and money—this question deserves careful consideration. If the decision is to continue with the solution, the next step, systems analysis, defines the problems and opportunities of the existing system. During systems investigation and analysis, as well as design maintenance and review, discussed next, the project must have the complete support of top-level managers and focus on developing systems that achieve business goals.

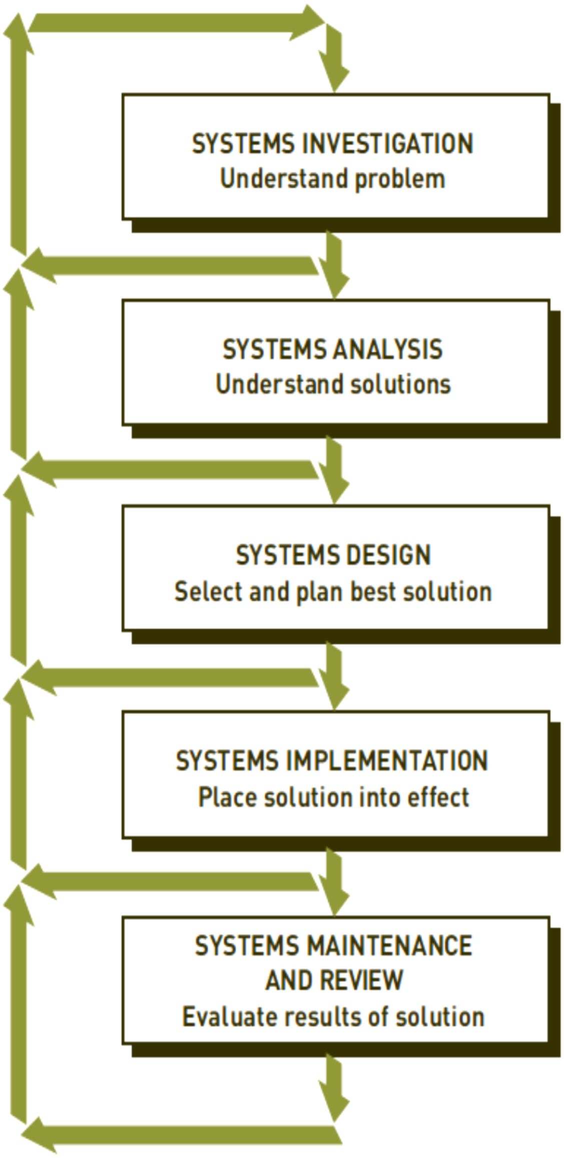


Figure 3: System Development Life

Systems Design, Implementation, and Maintenance and Review

Systems design determines how the new system will work to meet the business needs defined during systems analysis. Systems implementation involves creating or acquiring the various system components (hardware, software, databases, etc.) defined in the design step, assembling them, and putting the new system into operation. The purpose of systems maintenance and review is to check and modify the system so that it continues to meet changing business needs. Increasingly, companies are hiring outside companies to do their design, implementation, maintenance, and review functions.

Computer and Information Systems Literacy

Whatever your college major or career path, understanding computers and information systems will help you cope, adapt, and prosper in this challenging environment. Knowledge of information systems will help you make a significant contribution on the job. It will also help you advance in your chosen career or field. Managers are expected to identify opportunities to implement information systems to improve their business. They are also expected to lead IS projects in their areas of expertise. To meet these personal and organizational goals, you must acquire both computer literacy and information systems literacy.

Computer literacy is knowledge of computer systems and equipment and the ways they function. It stresses equipment and devices (hardware), programs and instructions (software), databases, and telecommunications. Information systems literacy goes beyond knowing the fundamentals of computer systems and equipment. Information systems literacy is the knowledge of how data and information are used by individuals, groups, and organizations. It includes knowledge of computer technology and the broader range of information systems. Most important, however, it encompasses how and why this technology is applied in business. Knowing about various types of hardware and software is an example of computer literacy. Knowing how to use hardware and software to increase profits, cut costs, improve productivity, and increase customer satisfaction is an example of information systems literacy.

Information systems literacy can involve recognizing how and why people (managers, employees, stockholders, and others) use information systems; being familiar with organizations, decision-making approaches, management levels, and information needs; and understanding how organizations can use computers and information systems to achieve their goals. Knowing how to deploy transaction processing, management information, decision support, and special-purpose systems to help an organization achieve its goals is a key aspect of information systems literacy.

GLOBAL CHALLENGES IN INFORMATION SYSTEMS

Changes in society as a result of increased international trade and cultural exchange, often called globalization, have always had a significant impact on organizations and their information systems.

Today, people in remote areas can use the Internet to compete with and contribute to other people, the largest corporations, and entire countries. These workers are empowered by high speed Internet access, making the world flatter. In the Globalization 3.0 era, designing a new airplane or computer can be separated into smaller subtasks and then completed by a person or small group that can do the best job. These workers can be located in India, China, Russia, Europe, and other areas of the world. The subtasks can then be combined or reassembled into the complete design. This approach can be used to prepare tax returns, diagnose a patient's medical condition, fix a broken computer, and many other tasks.

Today's information systems have led to greater globalization. High-speed Internet access and networks that can connect individuals and organizations around the world create more international opportunities. Global markets have expanded. People and companies can get products and services from around the world, instead of around the corner or across town.

These opportunities, however, introduce numerous obstacles and issues, including challenges involving culture, language, and many others.

- **Cultural challenges**. Countries and regional areas have their own cultures and customs that can significantly affect individuals and organizations involved in global trade.
- **Language challenges**. Language differences can make it difficult to translate exact meanings from one language to another.
- **Time and distance challenges.** Time and distance issues can be difficult to overcome for individuals and organizations involved with global trade in remote locations. Large time differences make it difficult to talk to people on the other side of the world. With long distance, it can take days to get a product, a critical part, or a piece of equipment from one location to another location.
- **Infrastructure challenges**. High-quality electricity and water might not be available in certain parts of the world. Telephone services, Internet connections, and skilled employees might be expensive or not readily available.
- **Currency challenges**. The value of different currencies can vary significantly over time, making international trade more difficult and complex.

- **Product and service challenges.** Traditional products that are physical or tangible, such as an automobile or bicycle can be difficult to deliver to the global market. However, electronic products (e-products) and electronic services (e-services) can be delivered to customers electronically, over the phone, through networks, through the Internet, or by other electronic means. Software, music, books, manuals, and advice can all be delivered globally and over the Internet.

- **Technology transfer issues.** Most governments don't allow certain military-related equipment and systems to be sold to some countries. Even so, some believe that foreign companies are stealing intellectual property, trade secrets, and copyrighted materials, and counterfeiting products and services.

- **State, regional, and national laws.** Each state, region, and country has a set of laws that must be obeyed by citizens and organizations operating in the country. These laws can deal with a variety of issues, including trade secrets, patents, copyrights, protection of personal or financial data, privacy, and much more. Laws restricting how data enters or exits a country are often called trans-border data-flow laws. Keeping track of these laws and incorporating them into the procedures and computer systems of multinational and transnational organizations can be very difficult and time consuming, requiring expert legal advice.

- **Trade agreements.** Countries often enter into trade agreements with each other. The North American Free Trade Agreement (NAFTA) and the Central American Free Trade Agreement (CAFTA) are examples. The European Union (EU) is another example of a group of countries with an international trade agreement. The EU is a collection of mostly European countries that have joined together for peace and prosperity.

Chapter 2

Introduction to Health Information Systems

Objectives

- Define the health information systems (HIS)
- Apply IT tools and approaches to health care field
- Discover what makes successful HIS
- Relate the use of data and enabling information technologies healthcare field

Healthcare information systems Definition

Healthcare information systems refer to such systems that are used to process data, information and knowledge in healthcare environments. While healthcare information systems and health information systems are often used today to refer to the same concept, a series of terms have been used in the evolution of this phenomenon from its early foundations in the 1960s. Though there is no clear consensus in literature until lately, the term health information systems is analogous to various primitive forms of this concept such as hospital information systems. Similarly, terms such as computerized patient records, electronic medical records, and the more current electronic health records have come to be commonly used almost interchangeably. Though the exact meanings may differ, all represent a progression in the development of healthcare information technology.

Why we need Health Information

Health information technology would allow medical professionals, such as doctors, nurses, physician's assistants, and other medical professional's easy access to a number of services, such as patient's records, dental services, nursing services, social services, and crisis and critical care services, etc. By using the health information technology to update and maintain information in real-time, available medical, social and community resources may be allocated appropriately and efficiently by the users of the technology. This may result in decreased costs for treatment of individual patients and for the health care system as a whole. Records of available services may be updated and stored in the database and accessed from multiple locations. Archival copies of the database may be analyzed over time to provide statistical data on the availability and effectiveness of medical and health services in the served geographic area. Clinical diagnoses and other data may be used for epidemiological analysis. For example, the PwC survey found that 60% of the population would be comfortable sharing data if they were used to coordinate care, and 54% would agree to share it if the data were used to support real-time decision-making for their care.

By analyzing the data collected by the technology, health care providers may improve health care services by sharing data across provider's boundaries. This will allow providers to improve services by ensuring the appropriate professional attends to a patient's individual needs and providing real-time information about the patient, care givers and available resources.

The technology may enhance an administrator's ability to examine trends in the allocation and utilization of nurses, social workers, doctors and any other health care providers in order to provide adequate staffing and staff availability. The technology could provide valuable information for both recruitment and retention of employees. The technology may also ensure that reliable data is available for doctors, nurses and social workers and provide data for statistical analysis. The technology may be implemented over a larger geographic area (such states) data from different geographic areas may be synthesized allowing the examination of trends from across the areas to aid in strategic decision making on many levels.

COMPONENTS OF A HEALTH INFORMATION SYSTEM

The Health Metrics Network's "Framework and Standards for Country Health Information Systems" describes the six components of a health information system and the standards needed for each. There is clear value in defining what constitutes a health information system and how its components interact with one another to produce better information for better decisions and better health. In addition to its six components, a health information system can be further divided into its inputs, processes, and outputs. Inputs refer to resources; processes touch on how indicators and data sources are selected and data are collected and managed. Outputs deal with the production, dissemination, and use of information. Accordingly, the six components of a health information system are as follows:

Inputs

1. **Health information system resources.** These consist of the legislative, regulatory, and planning frameworks required to ensure a fully functioning health information system, and the resources that are prerequisites for such a system to be functional. Such resources involve personnel, financing, logistics support, information and communications technology (ICT), and coordinating mechanisms within and among the six components

Processes

2. **Indicators.** A core set of indicators and related targets for the three domains of health information. Indicators need to encompass determinants of health; health system inputs, outputs, and outcomes; and health status.
3. **Data sources** can be divided into two main categories: (1) population-based approaches (censuses, civil registration, and population surveys) and (2) institution-based data (individual records, service records, and resource records). A number of other data-collection approaches and sources—occasional health surveys, research, and information produced by community based organizations—do not fit neatly into either of the two main categories but can provide important information that may not be available elsewhere.
4. **Data management.** This covers all aspects of data handling: collection, storage, quality-assurance, flow, processing, compilation, and analysis.

Outputs

5. **Information products.** Data must be transformed into information that will become the basis for evidence and knowledge to shape health action
6. **Dissemination and use.** The value of health information can be enhanced by making it readily accessible to decision makers (giving due attention to behavioral and organizational constraints) and by providing incentives for information use

For a health information system to function, policy, administrative, organizational, and financial prerequisites must be in place. Supportive legislative and regulatory environments are needed to enable confidentiality, security, ownership, sharing, retention, and destruction of data. Investment from domestic and international sources is required to strengthen ICT and provide human resources to run these systems. Expertise and leadership at national and subnational levels must also be provided to enable the monitoring of data quality and use. And infrastructure and policies must be in place to transfer information between producers and users both inside and outside the health system.

Limited national resources and capacities may affect the capacity of countries to apply the standards that the HMN framework proposes. Where standards are not in place, they are likely to evolve over time as countries adapt, use, and learn from the HMN framework.

Health Information System Developing Steps

(1) Review the existing system

(2) Define the data needs of relevant units within the health system

(3) Determine the most appropriate and effective data flow

(4) Design the data collection and reporting tools

(5) Develop the procedures and mechanisms for data processing

(6) Develop and implement a training programme for data providers and data users

(7) Pre-test, and if necessary, redesign the system for data collection, data flow, data processing and data utilization

(8) Monitor and evaluate the system

(9) Develop effective data dissemination and feedback mechanisms

(10) Enhance the HMIS

Healthcare information systems Trends

• **Trend 1: From Paper-Based Systems to Computer-Based Systems:** Meanwhile health data and information in the past have been created and stored mainly on paper; there has been a clear migration from paper to computer-based systems. This ability means that more data can be processed and stored through the use of modern information technologies to yield better knowledge. The future of healthcare information systems looks towards a near “paperless” era.

• **Trend 2: From Local to Global Information Systems:** While earlier healthcare information systems were limited to departmental units (e.g. radiology, or laboratory) or just within a healthcare practice system (e.g. hospital or clinic), modern healthcare systems target regional, national and even a global reach.

• **Trend 3: From Healthcare Professionals to Patients and Consumers:** Originally, healthcare information systems were designed to be used by mainly physicians and administrative staff (Ball, 1971; Ball et al., 1994), but it was later passed on to be used by nurses. Since then, the trend has shifted to involve more patient input.

• **Trend 4: From Using Data for Patient Care to Research:** Over the years, patient data has been used beyond patient care management to a more general use involving research in healthcare and even education.

• **Trend 5: From Technical to Strategic Information Management Orientation:** Computer supported information systems from the 1960s to the 1990s focused on problems resulting from the technical aspects of the systems, concerns about the organizational problems, social issues and change management aspects became more relevant at the turn of the millennium.

• **Trend 6: From Numeric Data to More Complex Forms of Data:** Not only has the technology that support *health information systems* advanced in technological complexity, the data that is being received and processed has also become complex. From numeric data through alphanumeric data to imaging and even molecular data

Health Information Systems Infrastructure and Information Flows

Health information technology consists of a wide range of networking technologies, clinical databases, electronic medical/health records, and other specific biomedical, administrative and financial technologies that generate, transmit and store healthcare information. In the diagram below, a generic model of information flows that typify health information systems infrastructure is presented, and a brief discussion of the application of this model is highlighted in Figure 1.

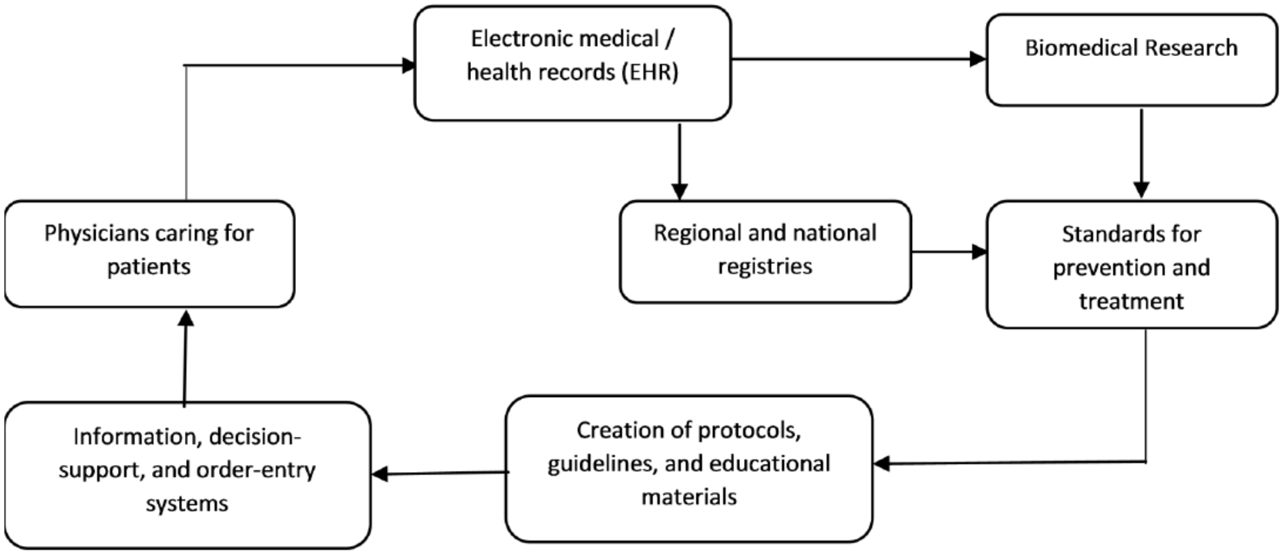


Figure 1: Information flows in a health information systems

In the model above, all information from healthcare providers (hospitals, clinics, emergency rooms, small offices, multispecialty groups, etc.) are entered into an electronic health record. This information is then networked to regional and national databases through electronic exchange. Data flows from EHRs and regional registries are then channeled into standards for prevention and treatment, which can be further processed to yield information for decision-making and decision-support. At each of these levels, appropriate information technologies are used to undergird data flow. The implications of this type of technological architecture are many-fold.

First, it raises issues of the encryption of data. The United States Health Insurance Portability Accountability Act (HIPAA) has set in place the privacy and security policies to provide guidance.

Second, the standards for data transmission and sharing over networks requires that all EHR developers all use the same standard—the HL7 standard. Third, given data transmission standards, data definition standards are equally important. They ensure that data communicated is read and understood by others. Fourth, with data coming from diverse healthcare sources, data quality control then becomes critical. Lastly, this model infrastructure means that regional and national databases with ability to hold, manipulate and produce useful information for decision-making.

Factors Influencing Successful Health Information Technology Implementation

There are no easy answers as to what contributes to successful HIT implementation projects. Reaping the benefits of electronic health records (EHRs) is in part dependent on successful implementation and implementation raises a host of technical, sociological and organizational issues that must be addressed before users are likely to accept the new system. Isolating the impacts of these factors is difficult although they can generally be classified into three broad categories: technical, sociological, and organizational. A few of the issues are noted below.

Technical. Is the information system intuitively easy to use? Is it easy to do the wrong thing within the system? Interoperability is another important consideration: Can the provider (physician's office or hospital) easily connect or share information from information-based systems that are purchased from multiple vendors? Does the information system support regulatory, accreditation, and legislative reporting requirements?

Sociological. Is the organization ready for the implementation? Do the implementers have the technical skills to install the system and have the users been trained to employ it effectively? Organizations cannot afford to have an unsuccessful implementation and the best guarantee against this is a clinical 'champion' (doctor, nurse, or other health professional) who has sufficient informatics training and education to lead the introduction of the EHR into practice.

Organizational. Does the facility or practice have access to skilled workers who understand the workflow of the organization and the potential limitations of HIT? Can these workers guide the organization's selection of information systems; integrate the new system(s) with existing systems and databases; train peers on using the systems; identify system limitations; and even help design next generation systems?

Chapter 3

Health Care Data Quality

Objectives

- To be able to discuss the relationship between health care data and health care information.
- To be able to identify problems associated with poor quality health care data.
- To be able to define the characteristics of data quality.
- To be able to discuss the challenges associated with measuring and ensuring health care data quality

Data Quality Overview

Accurate, timely and accessible health care data play a vital role in the planning, development and maintenance of health care services. Quality improvement and the timely dissemination of quality data are essential if health authorities wish to maintain health care at an optimal level.

In recent years, data quality has become an important issue, not only because of its importance in promoting high standards of patient care, but also because of its impact on government budgets for the maintenance of health services

The starting points for health care information are data and the collection of data, whether maintained manually or electronically at a large teaching hospital, health center or outlying clinic. Demographic and clinical data stored in a patient's medical/health record are the major source of health information and are of no value to medical science or health care management if they are not accurate, reliable and accessible.

Health care data

Health care data are items of knowledge about an individual patient or a group of patients. In health care, data are captured about a patient in paper or electronic format during his or her attendance at an outpatient clinic, community health center, primary health care provider, or his or her admission to a hospital. The data collected should include all relevant findings relating to the patient's condition, diagnoses, treatment, if any, and other events as they occur. Whether the data are collected manually or in a computer, it is important to ensure that the information is correct at the **point of entry**.

To ensure data quality, two key principles are **data accuracy and data validity**. To communicate effectively, data must be valid and conform to an expected range of values. To be useful, data must be accurate.

As the recording of data is subject to human error, there needs to be **built-in control measures** to eliminate errors, both in manual recording and computer entry.

Health Information

Health information is health care data that have been organized into a meaningful format. Health information may refer to **organized data** collected about an individual patient, or the summary of information about that patient's entire experience with his or her health care provider.

Health information can also be the **aggregate information** about all patients that have attended or been admitted to a hospital, or attended a health center, outlying clinic or a community immunization or health screening program.

Health information, therefore, can encompass the organization of a limitless array and combination of possible data items.

Health care information

Information should have value as a clinical review or management tool. Whether in a manual system or computer, information will not be valuable unless it is accurate, relevant, structured and presented in an easily useable form. Health care information should be capable of:

- promoting excellent clinical care;
- describing the types of individuals using services and the types of services they receive;
- measuring efficiency of the contact, treatment, referral and interaction by health care professionals;
- helping in the co-ordination of care between services provided;
- providing meaningful statistics for determining the health status of the community;
- measuring quality from a patient and provider perspective; and
- meeting accountability requirements.

Importance of data quality

Accurate and reliable health care data are needed for:

- determining the continuing and future care of a patient at all levels of health care;

- medico-legal purposes for the patient, the doctor and the health care service;
- maintaining accurate and reliable information about diseases treated and surgical
- procedures performed in a hospital and within a community, as well as immunization and screening programs, including the number and type of participants;
- clinical and health service research and outcomes of health care intervention, if required;
- accurate, reliable and complete statistical information about the uses of health care services within a community;
- teaching health care professionals; and
- working out staffing requirements and planning health care services.

Accurate and reliable health care data are used by:

- doctors, nurses and other health care professionals treating a patient admitted to a hospital or seen in an outpatient department or emergency room, and in community health centers, outlying clinics or general practitioners' offices. They use the medical/health record as a means of communication during an episode of care and treatment of a patient, and as an aide memoire for continuing care of that particular patient. Doctors also use health care data to evaluate the services provided;
- **nursing staff** in hospitals to evaluate data and develop critical pathways and patient care plans for admitted patients;
- **health insurers** who require information to reimburse the patient and/or health care facility for services rendered whether for an inpatient or ambulatory patient;
- **legal representatives and courts** as documentary evidence of a patient's care and treatment by a health care worker in a hospital, health center or clinic. They are also used to protect the legal interests of the patient, doctor and other health care professionals, the health care facility, and the public;
- **ministry of health** to review vital statistics and the incidence and prevalence of disease in a city, state or country. The provision of accurate and reliable aggregate data is important for public policy development and funding of health care services;
- quality assurance committees and medical staff as a basis for analysis, study and evaluation of the quality of health care services rendered to patients;
- researchers, to analyses and interpret data to determine causes, prevention methods and treatment for diseases, injuries and disabilities;
- health care facility accreditation and licensing agencies to review medical/health records to provide public assurance that quality health care is provided; and

- national governments who use the information to develop health care policy and provide and regulate funds.

Data quality — what it means

In general terms, quality, as defined by Donabedian (1988), consists of the ability to achieve desirable objectives using legitimate means. Quality data represent what was intended or defined by their official source, are objective, unbiased and comply with known standards. Data quality includes:

- **accuracy and validity** – of the original source data;
- **reliability** – data are consistent and information generated is understandable;
- **completeness** – all required data are present;
- **legibility** – data are readable;
- **currency and timeliness** – data are recorded at the time of observation;
- **accessibility** – data are available to authorized persons when and where needed;
- **meaning or usefulness** – information is pertinent and useful; and
- **confidentiality and security -** both important particularly to the patient and in legal matters.

Importance of data quality in health care

The quality of that data is crucial, not only for use in patient care, but also for monitoring the performance of the health service and employees. Data collected and presented must be accurate, complete, reliable, legible and accessible to authorized users if they are to meet the requirements of the patient, doctor and other health professionals, the health care facility, legal authorities, plus state, province and national government health authorities.

Components of data quality

Accuracy and validity

The original data must be accurate in order to be useful. If data are not accurate, then wrong impressions and information are being conveyed to the user. Documentation should reflect the event as it actually happened. Recording data is subject to human error and steps must be taken to ensure that errors do not occur or, if they do occur, are picked up immediately.

Example of accuracy and validity in a manual medical record system

- The patient's identification details are correct and uniquely identify the patient.
- All relevant facts pertaining to the episode of care are accurately recorded.

- All pages in the health record are for the same patient.
- The patient's address on the record is what the patient says it is.
- Documentation of clinical services in a hospital is of an acceptable predetermined value.
- The vital signs are what were originally recorded and are within acceptable value parameters, which have been predetermined and the entry meets this value.
- The abstracted data for indices, statistics and registries meet national and international standards and have been verified for accuracy.
- The codes used in hospitals to classify diseases and procedures conform to pre-determined coding standards.

To preserve data accuracy and validity in a manual system, processes need to be in place to monitor data entry and collection. In a computerized system, a computer can be instructed to check specific fields for validity and alert the user to a potential data collection error. In some instances, the computer does not allow an entry to be added if it fails the edit. In other instances, a warning is provided for the data entry operator to verify the accuracy of the information before entry.

Examples of edits and validity in a computer-based system

- In a hospital system, a patient must have a unique number because it is the key indexing or sorting field.
- The patient's number must fall within a certain range of numbers or the computer does not allow the data entry operator to move to the next field or to save the data.
- For hospital patients, the date of admission must be the same as or earlier than the date of discharge.
- A laboratory value must fall within a certain range of numbers or a validity check must be carried out.
- Format requirements such as the use of hyphens, dashes or leading zeros must be followed.
- Consistency edits can be developed to compare fields – for example a male patient cannot receive a pregnancy test.

Reliability

Data should yield the same results on repeated collection, processing, storing and display of information. That is, data should be consistent.

Examples of reliability

- The diagnosis recorded on the front sheet of the hospital medical record is consistent with the diagnosis recorded in the progress notes and other relevant parts of the medical record.
- Surgical procedures recorded on the front sheet of the hospital medical record are the same as recorded in operation reports in the body of the medical record.
- The age of the patient recorded on the first sheet of a medical/health record is the same as that recorded on other pages.
- The correct name of the patient is recorded on all forms within the medical/health record at the point of care or service given.

Completeness

All required data should be present and the medical/health record should contain all pertinent documents with complete and appropriate documentation.

Examples of completeness

- The first sheet of the medical/health record contains all the necessary identifying data to uniquely identify an individual patient.
- For inpatients, the medical record contains an accurately recorded main condition and other relevant diagnoses and procedures and the attending doctor's signature.
- Also for inpatients, all progress notes — from date of admission to discharge or death — are complete with signatures and date of entry.
- Nursing notes, including nursing plan, progress notes, blood pressure, temperature and other charts are complete with signatures and date of entry.
- For all medical/health records, relevant forms are complete, with signatures and date of attendance.

Legibility

All data whether written, transcribed and/or printed should be readable.

Examples of legibility

- Handwritten demographic data are clearly written and readable.
- Handwritten notes are clear, concise, readable and understandable.
- In all medical/health records, undecipherable codes or symbols are not used in either manual or electronic patient records.
- If abbreviations are used, they are standard and understood by all health care professionals involved in the service being provided to the patient.

Timeliness

Information, especially clinical information, should be documented as an event occurs, treatment is performed or results noted. Delaying documentation could cause information to be omitted and errors recorded.

Example of timeliness

- A patient's identifying information is recorded at the time of first attendance and is readily available to identify the patient at any given time.
- The patient's past medical history, a history of the present illness/problem as detailed by the patient, and results of physical examination, is recorded at the first attendance at a clinic or admission to hospital.
- On discharge or death of a patient in hospital, his or her medical records are processed and completed, coded and indexed within a specified time frame.
- Statistical reports are ready within a specified time frame, having been checked and verified.

Accessibility

All necessary data are available when needed for patient care and for all other official purposes. The value of accurately recorded data is lost if it is not accessible.

Examples of accessibility

- Medical/health records are available when and where needed at all times.
- Abstracted data are available for review when and where needed.
- In an electronic patient record system, clinical information is readily available when needed.
- Statistical reports are accessible when required for patient-care committees, planning meetings and government requirements.

Leadership in data quality

Many health care administrators already recognize that quality improvement is the way to add value to the services offered and that the dissemination of quality data is the only way to demonstrate that value to health care authorities and the community.

- **Health care administrators/managers** should be leaders in the move to improve the quality of data collected in the health care facility as they are responsible for the overall management of the facility and the quality of the information produced.
- **Senior doctors** should take the lead in ensuring data quality by taking time to ensure the more junior doctors record clinical data accurately and in a timely manner. Doctors should play an important role in maintaining data quality and should understand the need for accurate and timely data in the care of patients.

- **Senior staff** in departments such as **laboratory, radiology and pathology** and **community nurses, allied health professionals such as physical therapists, occupational therapists and social workers,** should be responsible for the provision of accurate and timely reporting and better checks on the quality of the content of reports.
- **Senior medical record staff** should ensure that the medical/health record is complete, available and accessible when needed.

Routine data quality monitoring

With standards in place, procedures relating to data collection and monitoring data quality should be carried out on a routine basis.

Two monitoring procedures for inpatients that have been undertaken for many years in some countries are quantitative and qualitative analysis of medical records.

Quantitative analysis of medical records

To evaluate the quality of documentation and, subsequently, patient care, the medical record must be complete. In a quantitative analysis, medical records should be reviewed to check that all documentation has been included (Huffman, 1963), for example:

- Patient identification is accurate and all details are complete.
- The history, physical examination, all progress and nursing notes are present, and all relevant reports such as pathology, X-rays, etc., are included.
- If the patient had surgery an operation and anesthetic report is present.
- All entries are signed and dated.

In other words, all relevant documentation must be present and authenticated.

Qualitative analysis of medical records

In a qualitative analysis of medical records, the information pertaining to patient care is reviewed for accuracy, validity and timeliness (Huffman, 1990). This includes:

- reviewing medical records to ensure that all clinically pertinent data have been accurately recorded; and
- Checking the front sheet to ensure that the patient's diagnosis and treatment have been recorded and are supported by documentation in the body of the medical record.

In an ideal situation, a staff member trained in quality assessment should perform a qualitative analysis on every medical record of discharged patients. This procedure takes a significant amount of time and in most situations, there is insufficient staff with the time available to complete the job effectively. The administrator or manager of a health care facility is generally responsible for determining what type of record to review and how often they should be reviewed.

In addition to the above, other procedures that could be implemented to ensure data quality are listed below.

Data entry checks

In a manual system, steps to check the entry of data in the medical/health record should be taken at the point of entry.

- Check the collection of demographic data by clerical staff in the admission office, emergency department and outpatient department reception, and health center and clinic reception/ registration area prior to the provision of health care services. The accuracy of this data is crucial for the identification of the patient during the present visit as well as their admission or future encounters with the health care service. Regular checks should be in place to prevent incorrect data being entered.
- Doctors and other health care professionals at all levels of health care should check the accuracy of the data for which he or she is responsible. In most medical/health records, data are recorded by a variety of persons, all of whom are responsible for the **accuracy of his or her documentation**. That is, the responsibility for **accurate and timely data entry rests with the professionals** involved and regular checks should be undertaken.
- When the medical/health record is returned to the medical record department (or the file room after attendance at a center or outlying post) after discharge or death or outpatient attendance, staffs are responsible for checking for completion before filing. They are responsible for monitoring the quality of the data and ensuring that health information generated from the medical record is timely, complete, accurate and accessible. The person responsible for the health record services, regardless of the type and level of health care, must manage those services in a manner that promotes quality information.

Checks on the quality of abstracted data

If a qualitative analysis is not undertaken on the complete medical/health record, a data quality check should be carried out on abstracted data. For most inpatient health care services, an abstract, that is the abstraction of information from a document to create a summary, is prepared at the end of a patient's hospital stay by the attending doctor. It is necessary to ensure the quality of the abstract information, for accuracy, validity, completeness and timeliness.

- A staff member, other than the initial clerk should **routinely check the abstracts.** To do this, the patient's record is retrieved from the database, or manually if a non-computerized record system is maintained, and the data elements are verified. In most cases, random samples are undertaken, errors noted are corrected and documented, and the staff member responsible for the original abstract re-trained.
- In a paper record, if a doctor forgets to sign an order, the order is not authenticated and cannot be carried out until the doctor signs.
- In a computer-based or electronic patient record, the authentication of an order is captured by a key word or code, and entered by the doctor when he or she has completed the order.

Medical/health record audit

Similar to a medical/health record review is a medical/health record audit, which is also a retrospective review of selected medical/health records or data documents to evaluate the quality of care or services provided compared with predetermined standards. To validly assess the completeness, accuracy, consistency, and legality of the medical record, an evaluation of the adequacy of medical record content can be conducted. Some steps identified by Jones (1993) following an audit of hospital medical records are listed below.

- The health facility administration and senior doctors should be asked to seek improvement in medical record documentation by assisting with development and design strategies to enhance data collection formats.
- Provision should be made for the allocation of sufficient resources to adequately monitor data quality.
- Support should be obtained from the administration to work with clinical departments and senior clinicians to examine strategies for the provision of adequate patient data.
- A comprehensive training program on documentation practices for junior doctors should be developed with the support of the hospital administration and senior medical staff.
- Continuous auditing of documentation practices should be carried out and findings monitored regularly.
- A multi-professional forum should be set up to address documentation and other issues and consider using a total quality management approach for improving the quality of data.

Development of an on-going quality assessment plan

If a health care facility or ministry of health is serious about data quality, they need to develop a plan aimed at improving and maintaining the quality of data and the information generated from that data.

To develop a quality plan, several structural components need to be in place. These include:

- commitment by top-level management to support the program, which would involve the appointment of a quality coordinator with adequate clerical support; and
- Staff responsible for quality control should be involved and deal with quality reports properly by reading and acting upon recommendations in a timely manner (Schofield, 1994).

To develop a data quality assessment plan, whether in a hospital, health center, clinic or aid post, the administration should take certain initial steps, which include:

- assign responsibility – a specific staff member should be assigned to audit aspects of documentation contained in the patient's medical/health record;
- identify important aspects of data collection – such as accuracy and validity, reliability, completeness and timeliness;
- determine indicators of data quality for each documentation component;
- set a threshold – that is, determine an acceptable error rate;
- **develop an organized method for collecting data** according to quality indicators previously developed;
- **assess actions** taken to improve documentation; and
- **Communicate the results** of the review/audit to those affected.

Quality assessment should be undertaken to ensure health information management functions are working effectively within the standards previously determined.

Performance improvement techniques

Along with a continuous quality assessment plan, steps should be taken to institute performance or quality improvement. This is a process by which a health care facility reviews its services to ensure quality.

- Staff responsible for health care services should be encouraged to not only meet a certain standard but should also seek to **improve their performance**.
- **Performance improvement** should not only include the staff of medical record services but also the entire staff of the facility and should be multi-disciplinary. To improve data quality, all persons involved in completing, checking and using data should be involved in insuring that the data are correct, valid, timely and relevant.

- Employees should use teamwork to improve a process. By instituting **performance improvement in data collection,** the outcome – patient care – will ultimately be improved.

Other steps to assist with data quality improvement

- **Performance indicators** can be developed as a guide to monitor and evaluate the quality and appropriateness of care. They are reliable and could be used to detect change such as health outcomes.
- **Policies** should be defined concerning the facility's overall position on quality. The policies should reflect a commitment to the highest standard of care for patients, including accurately and competently documented demographic and clinical information, and an opportunity for input to the program from all staff, with full support from the administration.
- A **quality review committee** should be established and charged with the responsibility of overseeing the quality activities program.
- A **quality coordinator** should be responsible for the day-to-day co-ordination of the program. This person must understand quality and its implementation and must be an effective communicator with the ability to impart knowledge to others. That is, a resource person who needs to network within and outside the organization.
- The **data collection system** needs to be **simple** and **user-friendly**.
- Quality activities need to **focus on practice** and not individual workers.
- **Confidentiality** needs to be maintained in all programs.
- **Staff education** requires that all staff clearly understand what quality means to the facility, how the program is managed, what is expected of staff and what they can expect to achieve.

Limitations in overcoming problems related to data quality

Data quality can be hampered by a number of issues, including the following.

- **Lack of uniformity of data** – without predetermined standards and uniform data sets, problems relating to the quality of health care data are difficult to solve.
- **Poorly designed data collection forms** – if forms are not well designed, the collection of data could be affected, resulting in poor quality data.
- **Limitations to doctors' capacity to communicate** – some doctors find it difficult to record data in a clear and concise manner, resulting in poor information. They also often use non-standard abbreviations and are "too busy" to complete medical records once the patient has been discharged from the facility or does not require further

treatment. Limited education in documentation requirements of medical staff is a major factor in poor data quality.

- **Limitations to information transfer from different parts of the facility** – sometimes information being transferred from the laboratory to the ward or a clinic does not contain the correct patient's name and medical/health record number. Such errors make it difficult to ensure that all data pertaining to an individual patient are filed in that patient's medical record. The transfer of information from one department to another or from a hospital to a clinic or aid post is often slow or information is lost in transmission.
- **Limited education of processing staff** – the processing of medical/health records requires staff that can understand the need for accuracy and completeness. If they are not properly trained, the production of quality data is threatened.
- **Lack of planning** by administrative staff to ensure data quality control programs are in place. All data collection and abstracting staff should be properly trained; and doctors should be educated in the requirements for accurate and timely documentation of patient care details.
- **No single record** – a problem of quality arises if more than one medical record is kept on each patient. Some facility staff, such as in cardiology, oncology and social work, insist on keeping their own records, thus limiting the overall collection of meaningful data about an individual patient.
- **Data discrepancies** – arising when errors occur at the point of collection and plans are not in place to check the entry and verify the data.

Chapter 4

History and evaluation of HIS

Objectives

- To be able to describe the history and evolution of health care information systems from the 1960s to the present.
- To be able to identify the major advances in information technology and significant federal initiatives that influenced the adoption of health care information systems.
- To be able to identify the major types of administrative and clinical information systems used in health care.
- To be able to discuss why information technology (IT) adoption rates are lower in health care compared with other industries.
- To be able to discuss the relationship between incentives and health care IT adoption and use.

HISTORY AND EVOLUTION

The history of the development and implementation of information systems in health care is most meaningful when considered in the context of a chronology of major health care sector and information technology events. In this section we explore the history and evolution of health care information systems in each of the past four decades and in the present era. We start with the 1960s and move forward to the current day

1960s: Billing Is the Center of the Universe; Managing the Money; Mainframes Roam the Planet

These early administrative and financial applications ran on large mainframe computers. Because the IS focus at the time was on automating manual administrative processes and computers were so expensive, only the largest, most complex tasks were candidates for mainframe computing. The high cost limited the development of departmental or clinical systems, although there were notable efforts in this direction, such as the Technicon system at El Camino Hospital. Most shared systems processed data in a central or regional data center. Like many of the in-house systems, most shared systems began with financial and patient accounting functions and gradually migrated toward clinical functions, or applications.

1970s: Clinical Departments Wake Up; Debut of the Minicomputer

By the 1970s, health care costs were escalating rapidly, partially due to high Medicare and Medicaid expenditures. Rapid inflation in the economy, expansion of hospital expenses and profits, and changes in medical care, including greater use of technology, medications, and conservative approaches to treatment also contributed to the spiraling health care costs. Health care organizations began to recognize the need for better access to clinical information for specific departments and for the facility as a whole. Departmental systems began to emerge as a way to improve productivity and capture charges and thereby maximize revenues. The development of departmental systems coincided with the availability of minicomputers. At the same time, improvements in handling clinical data and specimens often showed a direct impact on the quality of patient care because of faster turnaround of tests, more accurate results, and a reduction in the number of repeat procedures (Kennedy & Davis, 1992). The increased demand for patient-specific data coupled with the availability of relatively low-cost minicomputers opened a market for a host of new companies that wanted to develop applications for clinical departments, particularly turnkey systems. These software systems, which were developed by a vendor and installed on a hospital's computers, were known as turnkey systems because all a health care organization had to do was turn the system on and it was fully operational. Rarely could a turnkey system be modified to meet the unique information needs of an organization, however. What you saw was essentially what you got.

As in the 1960s, the health care executive's involvement in information system–related decisions was generally limited to working to secure the funds needed to acquire new information systems, although now executives were working with individual clinical as well as administrative departments on this issue. Most systems were still stand-alone and did not interface well with other administrative or clinical information systems in the organization.

1980s: Computers for the Masses; Age of the Cheap Machine; Arrival of the Computer Utility

Although the use of health care information systems in the 1970s could be considered an extension of the applications used in the 1960s with a slight increase in the use of clinical applications, the 1980s saw an entirely different story. Sweeping changes in how Medicare reimbursed hospitals and others for services, coupled with the advent of the microcomputer, radically changed how health care information systems were viewed and used. In 1982, Medicare shifted from a cost-based reimbursement system for hospitals to a prospective payment system based on diagnosis related groups (DRGs). This new payment system had a profound effect on hospital billing practices. Reimbursement amounts were now dependent on the patient's diagnosis and the accuracy of the ICD-9-CM codes used for each patient and his or her subsequent DRG assignment became critical. Hospitals received a predetermined amount based on the patient's DRG, regardless of the cost to treat that patient. The building and revenue enhancement mode of the 1960s and 1970s was no longer always the best strategy for a hospital financially. The incentives were now directed at ordering fewer diagnostic tests, performing fewer therapeutic procedures, and planning for the patient's discharge at the time of admission. Health care executives knew they needed to reduce expenses and maximize reimbursement. Services that had once been available only in hospitals now became more widespread in less resource-intensive outpatient settings and ambulatory surgery centers. As Medicare and many state Medicaid programs began to reimburse hospitals under the DRG-based system, many private insurance plans quickly followed suit. Hospitals were not the only ones singled out to contain health care costs. Overall health care costs in the 1980s rose by double the rate of inflation. Health insurance companies argued that the traditional fee-for-service method of payment to physicians failed to promote cost containment. Managed care plans began to emerge in parts of the nation, and they reimbursed physicians based on capitated or fixed rates.

At the same time, as changes were made in reimbursement practices, large corporations began to integrate the organizations making up the hospital system (previously a decentralized industry), enter many other health care–related businesses, and consolidate control. Overall there was a shift toward privatization and corporatization of health care. The integrated delivery system began to emerge, whereby health care organizations offered a spectrum of health care services, from ambulatory care to acute hospital care to long-term care and rehabilitation.

Although most organizations had patient demographic and insurance information available in their administrative applications, rarely were they able to integrate the clinical and the financial information needed to evaluate care and the cost of delivering that care in this new environment. Most of the clinical information systems or applications were being acquired piecemeal. For example, it was not uncommon for the director of laboratory services to go out and purchase from the vendor community the "best" laboratory information system, the pharmacy director to select the "best" pharmacy system, and so forth. This concept of selecting the "best of breed" among vendors and systems became prevalent in the 1980s and still exists to some extent today. Organizations that adopted the best-of-breed approach then faced a challenge when they tried to build interfaces or integrate data so that the different systems could interoperate, or communicate with each other. Even today, system integration remains a challenge for many health care organizations despite progress in the use of interoperability standards.

1990s: Health Care Reform Initiatives; Advent of the Internet

The 1990s marked another time of great change in health care. It also marked the evolution and widespread use of the Internet along with a new focus on electronic medical records.

The changes in physician reimbursement and the increased focus on prevention guidelines and disease management in the 1990s had implications for the community-based physician practice and its use of information systems. Up until this time, most of the major information systems development had occurred in hospitals. Some administrative information systems were used in physician practices for billing purposes, but as physician payment relied increasingly on documentation substantiated in the patient's record and as computers became more affordable, physicians began to recognize the need for timely, accurate, and complete financial and clinical information. Early adopters of clinical information systems also found electronic prompts and preventive health reminders helpful in managing patient care more effectively and efficiently. Likewise, more vendor products designed specifically with the physician practice setting in mind were becoming available.

In 1991, the Institute of Medicine (IOM) published its landmark report The Computer-Based Patient Record: An Essential Technology for Health Care. This report brought international attention to the numerous problems inherent in paper-based medical records and called for the adoption of the computer-based patient record (CPR) as the standard by the year 2001. At this point in the history and evolution of health care information systems, it is important to understand the IOM report's impact on the vendor community and health care organizations. Leading vendors and health care organizations saw this report as an impetus toward radically changing the ways in which patient information is managed and patient care is delivered. During the 1990s, a number of vendors developed CPR systems. Yet only 10 percent of hospitals and less than 15 percent of physician practices had implemented them by the end of the decade (Goldsmith, 2003). These percentages are particularly low when one considers the fact that by the late 1990s, CPR systems had reached the stage of reliability and technical maturity needed for widespread adoption in health care.

Five years after the IOM report advocating computer-based patient records was published, President Clinton signed into law the Health Insurance Portability and Accountability Act (HIPAA) of 1996. HIPAA was designed to make health insurance more affordable and accessible, but it also included important provisions to simplify administrative processes and to protect the confidentiality of personal health information. All of these initiatives were part of a larger health care reform effort and a federal interest in health care IT for purposes beyond reimbursement. Before HIPAA, it was not uncommon for health care organizations and health plans to use an array of systems to process and track patient bills and other information. Health care organizations provided services to patients with many different types of health insurance and had to spend considerable time and resources to make sure each claim contained the proper format, codes, and other details required by the insurer. Likewise, health plans spent time and resources to ensure their systems could handle transactions from a host of different health care organizations, providers, and clearinghouses.

The adoption of electronic transaction and code set standards and the greater use of standardized electronic transactions is expected to produce significant savings to the health care sector. In addition, the administrative simplification provisions led to the establishment of health privacy and security standards which went into effect in 2001 and 2003, respectively. It may be years before the full impact of HIPAA legislation on the health care sector is realized.

Health care organizations, providers, and patients could connect to the Internet and have access to a worldwide library of resources—and at times to patient-specific health information. In the early years of its use in health care, many health care organizations and vendors used the Internet to market their services, provide health information resources to consumers, and give clinicians access to the latest research and treatment findings. Other health care organizations saw Internet use as a strategy for changing how, where, and when they delivered health care services. The overall effects of Internet resources and capabilities on health care may not be fully realized for decades to come. We do know, however, that the Internet has provided affordable and nearly universal connectivity, enabling health care organizations, providers, and patients to connect to each other and the rest of the health care system. Along with the microcomputer, the Internet is perhaps the single greatest technological advancement in this era. It revolutionized the way that consumers, providers, and health care organizations access health information, communicate with each other, and conduct business.

With the advent of the Internet and the availability of microcomputers, came electronic mail (e-mail). Consumers began to use e-mail to communicate with colleagues, businesses, family, and friends. It substantially reduced or eliminated needs for telephone calls and regular mail. E-mail is fast, easy to use, and fairly widespread. Consumers soon discovered that they could not only search the Internet for the latest information on a particular condition but could then also e-mail that information or questions to their physicians.

The use of telemedicine and telehealth has also become more prevalent during the past few decades, particularly during the 1990s with its major advancements in telecommunications. Telemedicine is the use of telecommunications for the clinical care of patients and may involve various types of electronic delivery mechanisms. It is a tool that enables providers to deliver health care services to patients at distant locations. Most telemedicine programs have been pilot programs or demonstration projects that have not endured beyond the life of specific research and development funding initiatives. Reimbursement policies for these services vary, and that has been a significant limiting factor. In 2003, federal legislation allowed health care organizations to be reimbursed for professional consultations via telecommunication systems with specific clinicians when patients are seen at qualifying sites.

2000s: Health Care IT Arrives; Patients Take Center Stage

Health care quality and patient safety emerge as top priorities at the start of the millennium. In 2000, the IOM published the report To Err Is Human: Building a Safer Health Care System, which brought national attention to research estimating that 98,000 patients die each year due to medical errors.

Information is as critical to the provision of safe health care—care that is free of errors of both commission and omission—as it is to the safe operation of aircraft. To develop a treatment plan, a doctor must have access to complete patient information (e.g., diagnoses, medications, current test results, and available social supports) and to the most current science base [IOM Committee on Data Standards for Patient Safety, 2004].

Since the time the first IOM report was published, major purchasers of health care have taken a stand on improving the quality of care delivered in health care organizations across the nation by promoting the use of health care IT.

In addition to the considerable activity that has occurred at the national level in promoting adoption of health care IT and making price and quality information publicly available, significant technological advances have occurred in information technology. Electronic devices have become smaller, more portable, less expensive, and multipurpose. Broadband access to the Internet is widely available, even in remote, rural communities; wireless technology and portable devices (personal digital assistants, multipurpose cell phones, and so forth) are ubiquitous; significant progress has been made in the area of standards; podcasts, wikis, and Web 2.0 technologies have emerged; and radio-frequency identification devices (RFIDs), used more widely in other industries, have found their way into the health care marketplace. Consumers have also assumed a much more active role in managing their health and health information during the past decade by maintaining their own personal health records (PHRs). Unlike an EHR, which contains data collected and managed by a health care provider or organization, a PHR is consumer controlled. It is envisioned as a lifelong and comprehensive health record that is accessible from any place at any time. Health plans, insurance companies, and companies such as Google and Microsoft are making PHRs available to consumers, via secure Web sites, to store their personal health information. PHRs are described more fully in the next chapter, but they are introduced here as an important development in the evolution of health care information systems.

In the 2010s

Increased focus on value-based care as opposed to fee-based care and a drive to improve patient outcomes propel the growing accumulation of data to support clinical as well as operational decisions in health care.

Just as clinicians in the 1920s understood the importance of previous health records as learning tools that would improve outcomes, healthcare professionals leverage data to enhance care on a larger scale — using tools that analyze population health data.

New delivery models, such as accountable care organizations (ACOs), are implemented to contain costs, promote collaboration and improve patient health care. While ACOs, HIEs and growing health system networks have EHR and other systems to collect data, there is still a gap in aggregating and harmonizing the information from various systems to produce data that can be easily analyzed.

The Future

While there is no crystal ball to predict the future, it is safe to say that as health systems grow and expand to include other hospitals, physician practices and outpatient clinics, and as the volume of data grows with expansion, the need to integrate and harmonize data to make it available to all users is critical. Finding the right platform to support and enable access to structured and unstructured data across disparate systems is the first step to better preparing for a value-based future.

Interoperability, data-sharing and access to information will continue to be a critical requirement for process improvement, ACO enablement, information exchange and development of population-specific care that improve outcomes.

WHY HEALTH CARE LAGS IN IT

One might wonder why, with all the advances in information technology, the health care sector has been slow to adopt health care information systems, particularly clinical information systems. Other industries have automated their business processes and have used IT for years. The reasons for the slow adoption rate are varied and may not be readily apparent.

First, health care information is complex, unlike simple bank transactions, for example, and it can be difficult to structure. Health care information may include text, images, pictures, and other graphics. There is no simple standard operating procedure the provider can turn to for diagnosing, treating, and managing an individual patient's care. Although there are standards of care and practice guidelines, the individual provider still plays a pivotal role in conducting the physical examination, assessment, and history of the patient. The provider relies on prior knowledge and experience and may order a battery of tests and consult with colleagues before arriving at a diagnosis or an individualized treatment plan. Terminologies used to describe health information are also complex and are not used consistently among clinicians.

Second, health information is highly sensitive and personal. What could be more sensitive than a patient's personal health habits, family history, mental health, and sexual orientation? Yet such information may be relevant to the accurate diagnosis and treatment of the patient. Every patient must feel comfortable sharing such sensitive information with health care providers and confident that the information will be kept confidential and secure. Until HIPAA there were no federal laws that protected the confidentiality of all patient health information, and the state laws varied considerably. Today's younger generation, however, is very technologically savvy and far more comfortable with using the Internet for managing money, purchasing goods, seeking health information or second opinions, joining electronic support groups, and the like, so among these younger patients the concept of managing their own PHR may take off if the confidentiality and security of their health information can be assured.

Third, health care IT is expensive, and currently it is the health care provider or provider organization that bears the brunt of the cost for acquiring, maintaining, and supporting these systems. It has been very difficult to make a business case for the adoption of electronic medical records in small physician practices, where the bulk of health care is delivered.

Finally, the U.S. health care system is not a single system of care but rather a conglomeration of systems, including organizations in both the public and private sectors. Even within an individual health care organization there may be a number of fragmented systems and processes for managing information. Thus another major challenge facing health care is the integration of heterogeneous systems. Some connectivity problems stem from the fact that when microcomputers became available and affordable in the last half of the 1980s, many health care organizations acquired a variety of departmental clinical systems, with little regard for how they fit together in the larger context of the organization or enterprise. There was little emphasis initially on enterprise-wide systems or on answering such questions as, Will the departmental systems communicate with each other? With the patient registration system? With the patient accounting system? To what degree will these systems support the strategic goals of the organization? As health care organizations merged or were purchased from larger organizations, the problems with integrating systems multiplied.

Integration issues may be less of an issue when a health care organization acquires an enterprise-wide system from a single vendor or when the organization itself is a self-contained system. For example, Hospital Corporation of America (HCA), a for-profit health care system comprising hundreds of hospitals throughout the nation, has adopted an enterprise-wide system from a single vendor that is used across all HCA facilities. However, rarely does a single vendor offer all the applications and functionality needed by a health care organization. Significant progress has been made in terms of interoperability standards, yet much work remains.

Chapter 5

Architecture of a HIS

Objectives

- Define the information system Architecture
- Learn the HIS Architecture
- Learn the component of HIS

INFORMATION SYSTEMS ARCHITECTURE

An organization's information systems require that a series of core technologies *come together*, or work together as whole, to meet the IT goals of the organization. The way that core technologies, along with the application software, come together should be the result of decisions about what information systems are implemented and used within the organization and how they are implemented and used. For example, the electronic medical record system or the patient accounting system with which users ultimately interact involves not just the application software but also the network, servers, security systems, and so forth that all come together to make the system work effectively. This coming together should never be a haphazard process. It should be engineered. In discussing health care information system architecture, we will cover several topics:

- A definition of architecture
- Architecture perspectives
- Architecture examples
- Observations about architecture

A Definition of Architecture

Architecture is formally defined by IEEE as representing the fundamental organization of a system embodied in its components; their relationships to each other and to the environment; and the principles guiding its design and evolution. Our conceptualization of architecture, as the organization and evolution of a 'system of systems,' requires us to first understand the notion of a system. Very simply, a system represents a set of interacting or interdependent entities forming an integrated whole.

A system can be described as set of components that are interconnected through processes of input, throughput, output and feedback, it will suffice to say, that a system is not something which exists in objective reality, but represents our conceptualization of a certain phenomenon. For example, we may describe a hospital as a 'hospital system' because it consists of system with the characteristics of: **Input**: Unwell patients entering the hospital. **Throughput**: The different processes that the patient goes through in the hospital such as registration, billing, laboratory, Outpatient Department (OPD), Inpatient Department (IPD) and other more specialized ones such as blood bank and surgery. Output: Cured patient leaving the hospital (as one of the outputs). **Feedback**: This characteristic includes: *ƒ*. Direct feedback: Patients' opinion on the effectiveness of the hospital services, which may, then lead to either hiring of more specialist doctors (new inputs) or better management to improve the quality of services (throughput). Indirect feedback: The cure rates of the hospital is considered, which, if poorer than other hospitals, would lead to changes, such as, re-defining inputs (strengthening referral process to the hospital) or throughputs (establishing a stricter control on the quality and standards of care).

ƒ. While the above depiction of a system may seem rather simplistic and linear, it is done so deliberately for understanding. In practice, each part of the system can be expanded, and social dynamics added, to create a 'rich picture' of the system.

Cybernetic systems, based on feedback, are complex and full of unintended consequences shaping system behavior, and now being studied through theories like Complex Adaptive Systems. Building on this conceptualization of a system, an information system represents interconnected structures and processes, to enable the flows and use of information. In a broad sense, the term information system is frequently used to refer to the interaction between people, processes, data and technology.

The emphasis is not only on the Information and Communication Technologies (ICTs) that an organization uses, but also, on the way in which people interact with these ICTs and use information to support their ongoing business processes. While an information system is often assumed to be computer-based, it need not always be. Like, a manual flow of information (such as the movement of a paper file in an office) involving an input, throughput and output, can also be conceptualized as an information system. In the health system of most developing countries, major part of the data is still registered and reported using paper, but they nonetheless represent (health) information systems. HIS are quite simply defined as information systems in the health sector.

Architecture Perspectives

Organizations adopt various frames of reference as they approach the topic of architecture. This section will illustrate two approaches, one based on the characteristics and capabilities of the desired architecture and the other based on application integration.

Characteristics and Capabilities architecture as "the set of organizational, management, and technical strategies and tactics used to ensure that the organization's information systems have critical, organizationally defined characteristics and capabilities." For example, an organization can decide that it wants an information system that has characteristics such as being agile, efficient to support, and highly reliable. In addition, the organization can decide that its information systems should have capabilities such as being accessible by patients from their homes or being able to incorporate clinical decision support. If it wants high reliability, it will need to make decisions about fault-tolerant computers and network redundancy. If it wants users to be able to customize their clinical information screens, this will influence its choice of a clinical information system vendor. If it wants providers to be able to structure clinical documentation, it will need to make choices about natural language processing, voice recognition, and templates in its electronic medical record.

Application Integration Another way of looking at information systems architecture is to look at how applications are integrated across the organization. One often hears vendors talk about architectures such as best of breed, monolithic, and visual integration. *Best of breed* describes an architecture that allows each department to pick the best application it can find and that then attempts to integrate these applications by means of an interface engine that manages the transfer of data between these applications—for example, it can send a transaction with registration information on a new patient from the admitting system to the laboratory system. *Monolithic* describes the architecture of a set of applications that all come from one vendor and that all use a common database management system and common user interface.

Visual integration architecture wraps a common browser user interface around a set of diverse applications. This interface enables the user, for example, a physician, to use one set of screens to access clinical data even though those data may come from several different applications. This view of architecture is focused on the various approaches to the integration of applications; integration by sharing data between applications, integration by having all applications use one database, and integration by having an integrated access to data.

This view does not address other aspects of architecture: for example, the means by which the organization might get information to mobile workers.

Architecture Examples

A few examples will help to illustrate how architecture can guide information technology choices. Each example begins with an architecture statement and then shows some choices about core technologies and applications and the approach to implementing them that might result from this statement.

Statement: We would like to deliver an electronic medical record to our small physician practices that is inexpensive, reliable, and easy to support. To do this we will

- Run the application from our computer room, reducing the need for practice staff to manage their own servers and do tasks such as backups and applying application enhancements.
- Run several practices on one server to reduce the cost.
- Obtain a high-speed network connection, and a backup connection, from our local telephone company to provide good application performance and improve reliability.

Statement: We would like to have decision-support capabilities in our clinical information systems. To do this we will

- Purchase our applications from a vendor whose product includes a very robust rules engine.
- Make sure that the rules engine has the tools necessary to author new decision support and maintain existing clinical logic.
- Ensure that the clinical information systems use a single database with codified clinical data.

Statement: We want all of our systems to be easy and efficient to support. To do this we will

- Adopt industry standard technology, making it easier to hire support staff.
- Implement proven technology, technology that has had most of the bugs worked out.
- Purchase our application systems from one vendor, reducing the support problems and the finger-pointing that can occur between vendors when problems arise.

Observations about Architecture

Organizations will often bypass the architecture discussion in their haste to "get the IT show on the road and begin implementing stuff." Haste makes waste, as people say. It is terribly important to have thoughtful architecture discussions. There are many organizations, for example, that never took the time to develop thoughtful plans for integrating applications and that then discovered, after millions of dollars of IT investments, that this oversight meant that they could not integrate these applications or that the integration would be both expensive and limited. Organizations that have been very effective in their applications of IT over many years have had a significant focus on architecture. They have realized that thoughtful approaches to agility, cost efficiency, and reliability have a significant impact on their ability to continue to apply technology to improve organizational performance. For example, information systems that are not agile can be difficult (or impossible) to change as the organization's needs evolve. This ossification can strangle an organization's progress. In addition, information systems those have reliability problems can lead an organization to be hesitant to implement new, strategically important applications—how can they be sure that this new application will not go down too often and impair their operations?

Integrated Health Information Architecture

In the context of the health sector, there are multiple possible perspectives on how to 'circumscribe' the enterprise in question, when we discuss either a 'health system' more generally, or in a particular region/country, like the 'health system of India'. A hospital may be regarded as an enterprise or a business area, the system of district hospitals in a state, likewise, and the public health services in a state, may be regarded as an enterprise.

In fact, each business area, or health services area, within the health system may be de-limited and defined as an enterprise, within an 'enterprise architecture' framework. The overall health system 'enterprise architecture', is therefore seen consisting of a number of enterprise architectures, each of them dealing with a particular business area such as drugs, logistics, management, laboratories, HIV/AIDS anti-retroviral treatment, and hospitals. Likewise, enterprises can be defined in relation to the multitude of organizational units that make-up the health system (such as dispensaries, sub centers, primary health care centers and district hospitals). Other forms of enterprise or systems can be based on service functions and logistics (laboratories, drug supply, or ambulance services), or across various administrative and managerial levels (health facility, sub-district, district, or state). Regarded as an enterprise, the health system is made up of multiple enterprises, and even enterprises of enterprises, or system of systems.

If we include everything in an enterprise-based analytical framework, how is it possible then to handle the complexity? **Firstly**, identifying and defining the perspective on the enterprise, the business or functional area, on which to focus. **Second**; focusing on the information provided and its use to support management within the health system. This way, we can obscure irrelevant areas, such as the various production systems. However, achieving such clarity in practice is both technically and institutionally complex, for historical reasons.

Traditionally, each part, sector or program within the health sector has been developing their own information systems, tailored to serve their specific needs. These systems often are paper-based at the levels of data collection and reporting, and computerized at higher levels of the state/province or national levels. As there has been little or no coordination of these reporting systems and since most data originates from the local health services; health workers become overburdened by a plethora of reporting formats to fill in and report on every month. There are overlaps and inconsistencies between these reporting formats and the way data elements are named and defined, resulting in the reporting of the same data several times in different formats, and sometimes in different ways under multiple names. Quality of data and efficiency of the systems are adversely affected, leading to a vicious cycle of data not been used, because of its poor quality. And the less it is used, the more the quality suffers.

Architecture to Support Decision-making and Management

Fragmentation and lack of coordination of HIS have been identified by various researchers and also managers, as the major problem shaping their use and utility. Each health service, health program, project or initiative tends to organize their own reporting systems, often oblivious of what already exists, whether the data they require is already being collected under any other program or a different name?

Given that the main problems are fragmentation and lack of integration, how can then, a separate but still fragmented design and architectures for each of the subsystems improve the situation? The simple answer is that separate architectures for each sub-system will not necessarily lead to integration between the sub-systems. A general problem with information system design methodologies, which are based on mapping current work-flows and information handling practices, which is to some extent needed, is that they tend to focus on and conserve current practices and therefore do not necessarily enable innovation, that is support new ways to do the work, which new technologies necessarily enable.

Therefore, what is needed is, to first, take the perspective of the whole and overall health system as a point of departure, and second, replicate this perspective at each level of the health services; from the national and state levels to the levels of district, sub-district, and health facilities. What is common for each administrative level is the need for information to inform decision-making and to support management. Key indicators and information more generally related to public health and health management at the particular level in question is in contrast to clinical information related to individual patients. This information by definition needs to encompass at least the scope of management and decision-making, meaning that key information from all sub-areas are needed. While the national and state levels will mostly deal with policy-making and evaluation, the district is responsible for the operational management of health services delivery, including vertical health programs in the district. This requires more of monitoring than evaluation related information. For example, while at the national level, one needs to know the overall state of immunization coverage, to be able to evaluate the effectiveness of the immunization program; at the district level, one needs more detailed monitoring information such as information related to drop outs and vaccine supply and so on. Further, these services are implemented and delivered by various health facilities in the district. Translating this into the language of ‘enterprise architecture’, we may say that each business area identified for the architecture is that of management, co-ordination and decision-making. An IHIA should then be designed to meet these cross-cutting information needs.

The rapid developments of Internet and mobile infrastructure have led to new computerized and mobile-based information systems that have been planned for and implemented. The worry is, of course, that the current process of computerization only replicates the former situation of fragmentation and poor co-ordination, though not casting it in stone, but wiring it up in the computer infrastructure. To guard against this, we propose an approach to design, based on information use.

At all levels of management, for supporting processes of co-ordination and decision-making, key information is needed. This support could be leveraged from various data sources, including routine data collection, and other relevant areas; for example, from the census data and population based surveys of health status and utilization of services. The approach to focus on information use, shares the generic characteristic of information for decision support across all administrative levels based on available and relevant data sources. This provides for the foundation, and a replication of design processes, both vertically (across administrative levels) and horizontally (across programs at different levels) to establish the systems of systems – or in our words the 'IHIAs'.

Service Oriented architecture for healthcare

Service Oriented Architecture (SOA) generally, health information is stored over a number of different WSP. A national Power system must be available for the provision of directory services to determine the distributed locations of the source systems holding the related health records. A model supports secure communications between healthcare providers and the Power System in the national e-health environment as shown in Figure 1.

These services can be performed at the level of the WSP. The access control and authorization process is best performed close to where the source system is, as each healthcare service provider might implement the service differently based on its own WSP access requirements. There are no centralized network provisions to handle peer-to-peer communications; each service must manage its own interface to the network.

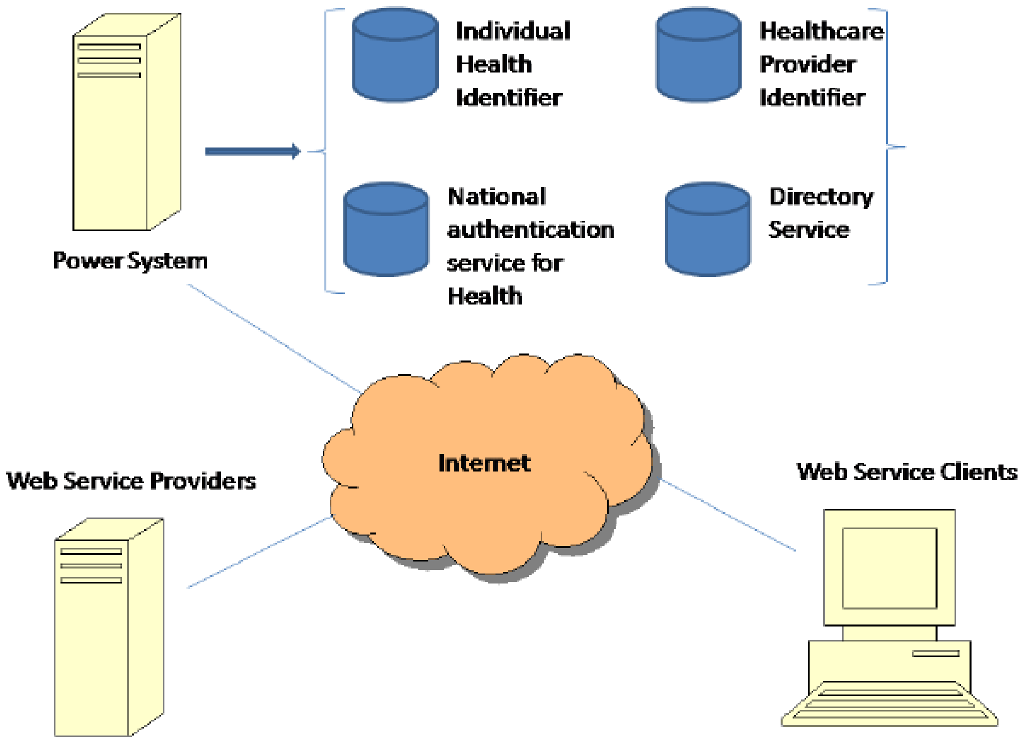


Figure 1 Service Oriented Architecture (SOA)

Chapter 6

HIS Opportunities and Challenges

Objectives

- State the various types of Opportunities and challenges
- Analyze the Opportunities and challenges of HIS
- Discuss how to get the benefit of the opportunities and override the challenges

OPPORTUNITIES

Healthcare information systems have been critically acclaimed for their ability to increase legibility, reduce medical errors, shrink costs and boost the quality of healthcare. In the following subsections, the potential opportunities that lie in HIS are examined.

Cost Savings

Health information systems are expected to save money in the long run and generate organizational profitability through efficiencies, cost-effectiveness and safety of medical deliveries. Practically-speaking, it is expected that HIS will reduce expenses associated with record-keeping while meeting privacy regulation standards and improving workflows, practice management and billing. HIS is also expected to permit automated sharing of information among providers, reduce office visits (to receive tests results) and hospital admissions (due to missing information), and even reduce risks of malpractice law suits. The information technology (IT) investments in the healthcare industry leads to increased profitability and quality products and services.

Healthcare Information record (EMR) systems were effectively and sufficiently implemented. With the adoption and implementation on interoperable EMR systems they were even more optimistic, estimating a cumulative net savings totaling another $142-$371 billion over a 15-year period. There seems to be no question that long-term savings is a potential economic strength of health IT systems.

Reduction in Medical Errors

The Institute of Medicine (IOM) (1999) study reported that up to 98,000 people die in U.S. hospitals each year as a result of preventable medical errors alone. It further predicted that 50% of errors could be eliminated over a five-year period if existing technological know-how was implemented. A more recent report noted yearly increases in medical errors—claiming a disturbing 1.5 million adverse drug events due to preventable medical errors. In its list of solutions to this problem, the IOM unequivocally mentioned the use of health information technologies—such as e-prescription—as a key solution element. Evidently, health information systems' role in increasing legibility and medical error reduction in healthcare services has been shown to be a potential benefit.

Overall Quality of Healthcare

While reduction in errors certainly contributes to the quality of healthcare, there are more general opportunities that HIS offer. HIS has many contributes to improve overall quality of care and patient outcomes in a population. These include:

• More complete, accurate and structured clinical data documentation;

• Automatic sorting and summarization of data for information generation;

• Direct access to instant updates to records as well as remote access to patient records;

• Reduced medical mistakes from legibility and order entry errors;

• Increased decision support from structured data and predictive modeling and disease management tools;

• Data mining capabilities provided by the vast amounts of structured medical record data contributing to disease research and preventive interventions in clinical care; and

• Continuous improvement in clinical decision making through decision support (enabled by health information exchange), rapid dissemination of information and quicker monitoring of care. Through the aforementioned capabilities of HIS, mistakes are kept at bay, information quality is enhanced, treatment response times are improved, and optimal decision-making is attained.

CHALLENGES

In spite of the huge potential and opportunities that lie in HIS to radically transform healthcare and the healthcare sector, many challenges are evident and imminent. The adoption of IT in healthcare has been particularly slow and lagging behind that of major industries by as much as 10-15 years. This is further exacerbated by the failure in HIS implementation as well as resistance to the use of the technology by healthcare professionals

These challenges range from issues related to the technology itself, the healthcare setting, system users and the regulatory environment. For instance, Blumenthal (2009) lists the barriers faced by healthcare information technology proponents in the U.S. namely: *low adoption rates by doctors and hospitals due to associated costs, perceived lack of return on investments, use issues and concerns of privacy and security*. Generally-speaking, challenges stem from the interaction of technical, human, and organizational factors affecting the adoption and use of these healthcare systems. To better explain these factors in their proper contexts.

The success/failure of an HIS is contingent on the gap between "design conceptions of HIS" and the "current realities." This paradigm argues that the two major stakeholders of HIS, namely system designers and the system users both possess their different but subjective versions of reality. Furthermore, because these groups are especially valuable to, and different from each other, their interaction produces the challenges that HIS faces.

More specifically, the "design-gap" framework provides a lens for pitching HIS designers' view of the technology and its context, versus HIS end-users' view of the same technology. Based on this, the "design reality" gap paradigm presents three archetypes of hard-soft gaps that are crucial to understanding healthcare challenges. These archetypes are technical rationality, managerial rationality and medical rationality.

Technical rationality: Technical rationality depicts a technology-based worldview where everything is supposed to be objective and rational—not subject to personal, cultural, and political influences. Designers of HIS technology are typically dominated by IT professionals with this kind of mindset. They design a system with the view that is would be looked upon rationally and objectively. Hence, they emphasize on the specifications and the technical designs that will yield particular outcomes.

Nevertheless, technical challenges still arise from lack of standardization of technology, the absence of a well-developed healthcare information exchange (HIE) which will permit healthcare institutions in a given region to be able to freely share healthcare data.

The ability to have an interoperable health information exchange that can both share information quickly and seamlessly also raises concerns on privacy and security of electronically transmitted data. We observe therefore that, even though these systems are mostly built from a technical worldview point, issues on usability, standardization and interoperability further exacerbate the problem.

Managerial rationality: A managerial worldview of HIS emphasizes the economic and socio-political outlook of systems. Typically, managers are concerned about the costs, return on investments, and even the interest of external stakeholders like the government. They perceive the system from the standpoint of the surrounding socio-political and economic system within which the technology is supposed to be embedded. Like technology, money is usually considered as a rational entity. When financial information is perceived to have a role in HIS, those information systems are likely to be viewed through an objective and rational model. This is particularly true when a finance-based worldview dominates design inscriptions.

HIS direct and indirect costs remain a major concern of many healthcare institutions. This is particularly so, because of the high initial investments and the low perceived return on investments. Also it is noted that business process re-engineering is also a difficult issue to handle. Most changes that come with HIS implementation require huge organizational changes requiring not only financial investments but a total change in the way business is conducted. Lastly, there exists interdependence between financial and clinical outcomes that dictates to a reasonable extent how much investment should be made to achieve a particular health outcome. Hence, cost of acquisition, running and maintenance of HIS is still a veritable barrier.

Medical rationality: Though this dimension focuses primarily on medical personnel, it is also considered in an objective and rational sense when diseases and injuries (but not patients) are the focal entity. When medical information is seen to play a central role in HIS, these information systems are therefore themselves likely to be conceived according to an objective and rational model. This would be the case in a design where clinicians or other healthcare professionals dominate the design process causing a medicine-based worldview to prevail in design inscriptions. Medical rationality is likely to explain the wide and massive resistance to HIS since its inception. Physicians and other healthcare personnel view the system from an entirely different paradigm than IT personnel or managers.

The area of end-user resistance to information technology is clearly rising, but researchers will need to know how and why resistance to information systems occurs, especially in HIS environment. Additionally, training of dedicated health personnel to support HIS implementation and meet the standards of anticipated healthcare outcomes is critical. Hence, a clear challenge in HIS adoption is end-user resistance to the systems, as well as the lack of dedicated practitioners.

Chapter 7

Types of HISs

Objectives

- State the various types of HIS
- Compare the different types of HIS
- Choose the suitable Health Information System.

Introduction

As we mention in chapter 1 any Information system consist of three main component; Input, Process, and output. A system receives data - inputs which is then possibly processed in some way before producing some type of output. Depending upon our viewpoint we may focus on the input, processes or output. It is often felt that certain models of systems consider in too greater depth the input aspects to the detriment of other aspects such as the processes and more importantly the output. Because of this, 'output based' specifications have become popular where the model concentrates on describing the present or required outputs.

So the health information system can be categorical according to some diagonals:

- Subject and Task based systems
- Operational/tactical and strategic Health information Systems
- Clinical and Administrative Health information Systems
- The Electronic Health/Patient Record (EHR / EPR)
- Financial and Clinical Health Information Systems
- Decision Support Systems (DSS)
- Robotics and Simulators
- Telemedicine, Telematics and eHealth Systems
- Computer Simulations

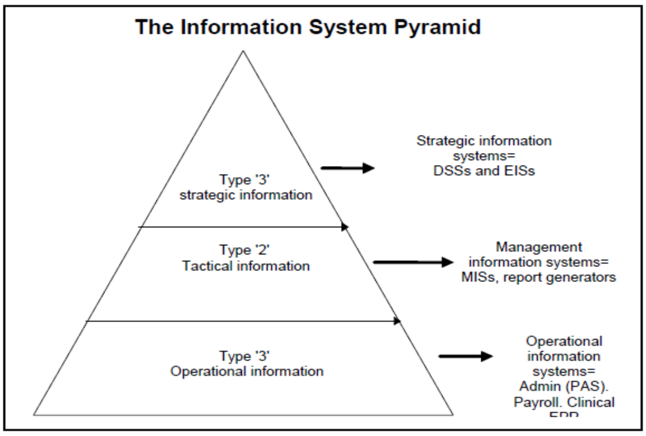
Subject and Task based systems

James Martin, as long ago as 1981, suggested that you could divide information systems into those that are either 'subject' or 'task' based, although he did not use these exact terms. He defined a 'subject' based system to be one which related to a particular thing in the organization such as a patient or doctor. In contrast a task based system was one that supported a particular task. Examples of task based systems would be standalone operating theatre or admissions/ discharge systems.

He suggested that 'subject' rather than 'task' based systems were best. The reason for his preference is that it reduces data duplication. In a task based system if a subject often undergoes many tasks, basic details (e.g. name and address) would be collected each time, in contrast in a subject oriented system basic information would be collected once and would flow from task to task.

Operational/tactical and strategic Health information Systems

One of the most common ways to classify information is to use the operational, tactical, strategic divisions. At each level of the information pyramid there are also information systems which deal specifically with that type of information.

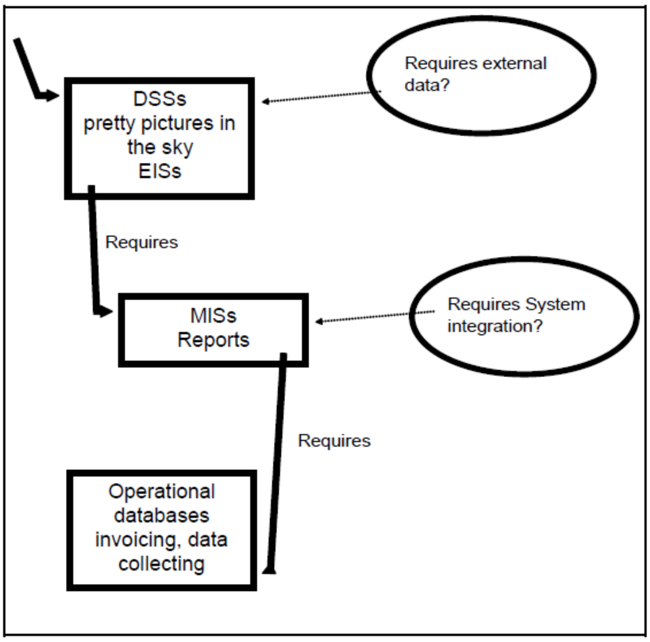


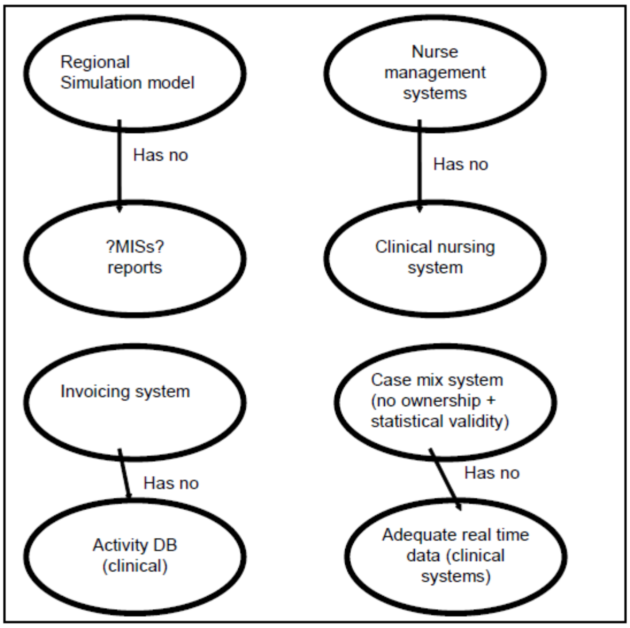
AbouZahr C, Adjei S, Kanchanachitra C, 2007 From data to policy: good practices and cautionary tales Lancet 369: 1039–46

The pyramid classification has several advantages:

- It allows assessment of how far down the road of computerization an organization is. This can be done because operational systems are usually developed before MISs (Management Information Systems) or EISs (executive Information Systems).
- It allows the highlighting of any uneven or inappropriate systems development. This is by considering the hierarchical data dependency, management information systems requiring an operational system to feed them.

By considering the dependencies illustrated on the left hand side one can identify deficiencies in individual systems as given on the right hand side. Examples of using this approach are given below.





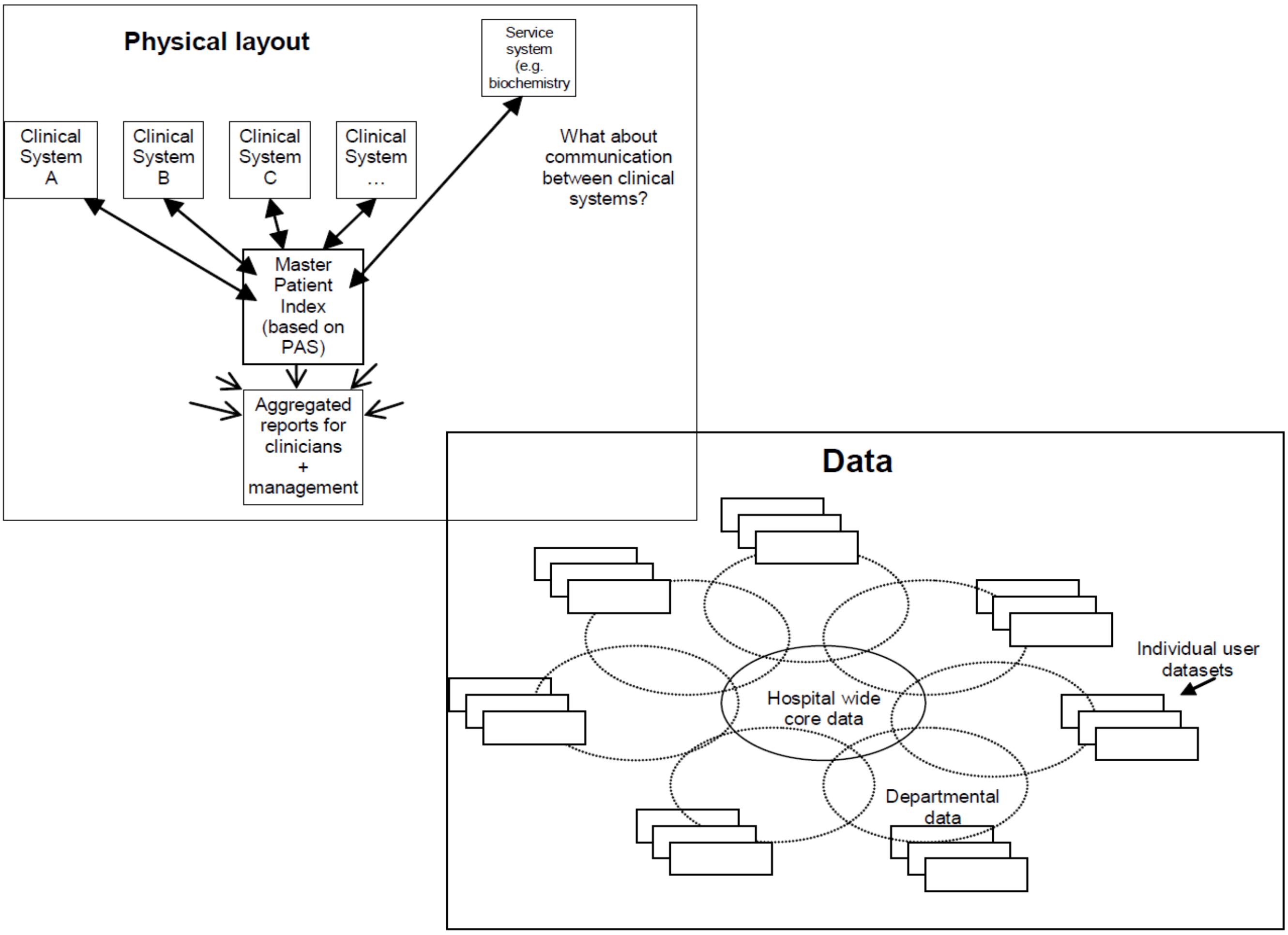
1. The former Northern Regional HA (UK) developed a computerized planning tool (a simulation) which provided output information concerning possible future hospital requirements by projecting hospital capacities and waiting list information. However this did not have the necessary feeder systems to keep it up to date.
2. All hospitals have problems working out costings which had to be done by top down apportioning as there are no feeder systems providing data on actual usage per client. In contrast American hospitals frequently use item billing systems.
3. Several hospitals have nurse management systems the data for which is gathered manually by collecting a plethora of data on paper, much of which could be obtained from a clinical system directly.

Clinical and Administrative Health information Systems

Another division that is often made is that between clinical and administrative systems. Yet if one considers it is basically impossible to develop any clinical system without it depending on some type of administrative data. For example the most basic of clinical systems should allow the production of letters to GPs or patients for follow-up requiring GP and address details. The question is do such details constitute administrative or clinical information?

Considered rather simplistically the core of an integrated hospital clinical information system is nothing more than a 'master index' consisting of the most basic of patient details ('administrative information') providing links to various clinical systems. Each departmental clinical system then allows individuals to set up additional 'research dataset's for specific activities. One can argue that each clinical system contains an Electronic Patient Record (EPR) or the virtual joining of each together for a specific patient represents an EPR.

This is probably defined as an administrative system because it was designed to allow retrospective data entry (i.e. information about the patient was usually entered after discharge when the notes get to medical records) and provided details of each 'episode' of care. Yet it is interesting to note that the reports, with the minimum of change had been called contracting datasets. Similarly the dataset also contains information about diagnosis, procedures and outcomes, all of which could be classed as clinical.



6. The Electronic Health/Patient Record (EHR / EPR)

A separate chapter describes this concept in detail. There are various standards being developed such as the EU standard for the above called the Electronic HealthCare record (**EHCR**). However more excitingly there is also the development of an open (i.e. free) standard, called the openEHR.

OpenEHR is a set of open specifications for Electronic Health Record (EHR) architecture – but it is not a software application. Its design purpose is to enable semantic interoperability of health information between, and within, EHR systems – all in a non-proprietary format, avoiding vendor lock-in of data.

7. Financial and Clinical Health Information Systems

Another division is often made between financial and clinical systems but once again it is easy to see that patient costing, if carried out on a patient usage basis is really tagging the various items used (as would be recorded in a clinical system) with a price. However, most costing is carried out on an estimated basis based upon apportioning the total costs retrospectively or more frequently on past years costs. One important aspect of financial systems is that of invoicing and non-payment follow-up.

8. Decision Support Systems (DSS)

'Decision support' is a phrase that has been bandied around for some time now and is usually linked with AI (Artificial Intelligence). Basically getting the computer to attempt to carry out some of the processing that the user does when converting the data ('facts') into information ('clinically relevancy'). While the technical abilities to develop DSS's in healthcare has been possible for well over a decade now few have been taken up to any significant extent for 'professional' organizational reasons.

Most people consider a decision support system to offer one of three levels of support:

- Presents the data in a way conducive to cognitive processing by sorting, classifying, flagging etc. Thus facilitating decision making by the user. For example presenting a list of drugs for asthma rather than just a list of drugs for all conditions.
- Provides the results of some data manipulation. Here the system mimics part of the cognitive process e.g. provides a list of drugs only suitable to treat Asthma in an 8 year old who has no other illness.
- Provides the results of some data manipulation and carries out some appropriate action. Here the system mimics more of the cognitive process as well as the output processes e.g. system prescribes drug and arranges next appropriate appointment.

There are social implications of adopting any of these three levels

A large number of applications (pieces of software) can be considered to be 'decision support systems' at the lowest level described above. A reference manager, electronic diary, statistical package and an online library catalogue all fulfil the criteria, and incidentally are all databases. In contrast both the Internet, without some type of filter, and a word processor are not.

9. Robotics and Simulators

Medical robotics is becoming an ever increasingly important part of surgery. Simulators are more prevalent in healthcare education. The main company who produce Resusci Anne has now branched out to develop various other simulators such as SimMan which includes complex software to mimic various cardiovascular parameters. Other simulators exist of more advanced clinical training including dentistry and colonoscopy

10. Telemedicine, Telematics and eHealth Systems

Health Telematics systems are another type of information system. Telematics is the electronic transfer of complex data from one place to another. Usually the data is video or multimedia. Therefore teleconferencing (having conferences by video link) is a type of telematics. A common method nowadays is to use Web based technology.

Telematics is being used increasingly in areas of the developing world where expert resources are scarce or the population density is low. Examples are sending Images such as X-rays to be reported upon remotely; often in another country such as hospitals in Denmark sending pictures to reporters in Lithuania

Various varieties of Tele-monitoring are also emerging, from remotely providing medial support to special care baby units to social care for the elderly.

The greatest change in the last few years, from the perspective of the patient (client / consumer etc.), is the use of the web, you now have 24 hour free access to doctors and various therapies including cognitive behavioral therapy

Health related activities that take place on the web, or facilitated by it, usually come under the title of eHealth, but once again this title has no universal definition. The problems with many telematics/eHealth projects are that they tend to be technology driven rather than demand driven from the clinical perspective. Jeremy Wyatt (Aberdeen) has written very sensibly on such issues.

11. Computer Simulations

Computer Simulations (in contrast to simulators used for teaching) are pieces of software ('applications') that allow you to create and manipulate a particular model. For example the games sim-city and sim-earth allow you to create cities or a whole world respectively.

Simulation is the most recent of the methods available to develop planning estimates, primarily because large simulations require powerful computers (either by using a super computer or by developing a virtual network. In contrast to the old mathematical methods the user is not limited to any assumptions inherent in the technique. Models of any level of sophistication can now be built with relative ease given the necessary resources and data.

In the health service relatively few simulations have been built. Early examples in the UK include: The Hermes project to enable managers to predict case mix changes in acute provider units. Other types of simulations tend to concentrate of specific problems such as costs of treating depression. In contrast others take a more epidemiological approach looking at the cost benefits of Influenza vaccination.

In America there are now Health care simulation consultancies providing advice to major health care organizations. There is also specific healthcare simulation software providing a point and click interface, which you can buy to develop models, such as Medmodel

In the last few years the development of social networking sites and virtual reality environments such as secondLife, means that a whole new environment is about to be unleashed and will probably have major consequences for Health Information Systems.

Chapter 8

Technology that support HIS

Objectives

- To gain a basic understanding of the core technologies behind health care information systems
- To be able to discuss emerging trends in information technology (such as mobility, Web services, Internet, wireless).
- To be able to identify some of the major issues in the adoption of information technologies in health care organizations.
- To be able to discuss why it is important for a health care organization to adopt an overall information systems architecture.

Overview

Although we do not believe that health care executives need to become information technology (IT) experts in order to make informed decisions about which health care information systems to employ in their organizations, we do believe that an exposure to some of the core technologies used to develop and implement common health care information systems is quite useful. This knowledge will help health care executives be more informed decision makers.

This chapter provides a broad view of several categories of core, or base, technologies. They are not unique to health care but are frequently found in health care organizations. We discuss technologies used in each of the following categories:

- Data management and access
- Networks and data communications
- Information processing distribution schemes
- Clinical and managerial decision support
- Trends in user interactions with systems

DATA MANAGEMENT AND ACCESS

All the health care applications discussed thus far require data. The electronic medical record (EMR) system relies on comprehensive databases, as do other clinical applications. Data must be stored and maintained so that they can be retrieved and used within these applications. In this section we discuss common types of databases and the database management systems with which they are associated. The majority of our discussion centers on the *relational database* because it is the type of database most commonly developed today. Two older types of databases, hierarchical and network (not to be confused with a computer network), may still exist in health care organizations as components of older, legacy applications, but because they no longer have a significant presence in the database market, they are not discussed here. A fourth type of database, the *object-oriented database*, has received a lot of attention in the literature during the past few years. Although a "pure" object-oriented database is not yet common in the health care market, there are applications with object-oriented components built upon relational databases. This hybrid database type is referred to as an *object-relational database*.

Relational Databases

Today the relational database is the predominant type used in health care and business. A relational database is implemented through a *relational database management system* (RDBMS). Microsoft Access is an example of an RDBMS for desktop computing; Oracle, Sybase, and Microsoft SQL Server are examples of the more robust RDBMS that are used to develop larger applications.

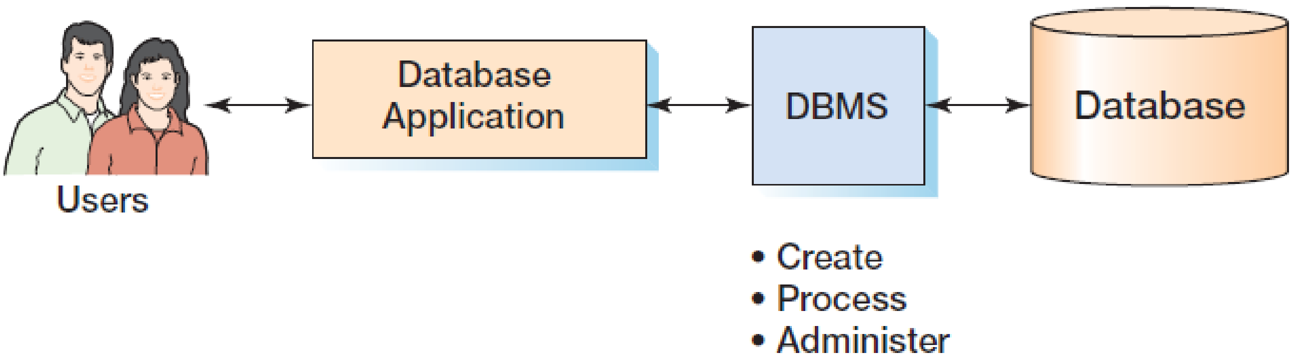


Figure 1: The Components of a Database System

As shown in Figure 1, a **database system** is typically defined to consist of four components: users, the database application, the database management system (DBMS), and the database. However, **Structured Query Language (SQL),** an internationally recognized standard language that is understood by all commercial DBMS products. **The database** is a collection of related tables and other structures. The database management system (DBMS) is a computer program used to create, process, and administer the database. The DBMS receives requests encoded in

SQL and translates those requests into actions on the database. The DBMS is a large, complicated program that is licensed from a software vendor; companies almost never write their own DBMS programs. **A database application** is a set of one or more computer programs that serves as an intermediary between the user and the DBMS. Application programs read or modify database data by sending SQL statements to the DBMS. Application programs also present data to users in the format of forms and reports. Application programs can be acquired from software vendors, and they are also frequently written in-house. The knowledge you gain from this text will help you write database applications.

Users, the fourth component of a database system, employ a database application to keep track of things. They use forms to read, enter, and query data, and they produce reports to convey information.

RELATIONAL DATA MODELING

Figure 2 is an example of an *entity relationship diagram* (ERD), which graphically depicts the tables and relationships in a simple relational database. Data modeling is an important tool for database designers. Although a complete discussion of PERSPECTIVE ERDs (and data modeling in general) is beyond the scope of this book, we will point out several key components here because these models are frequently used not only as ''blueprints'' for building databases but also as tools for communication between the designers and the eventual users. Therefore it may be necessary for the health care executive to have a cursory understanding of their components.

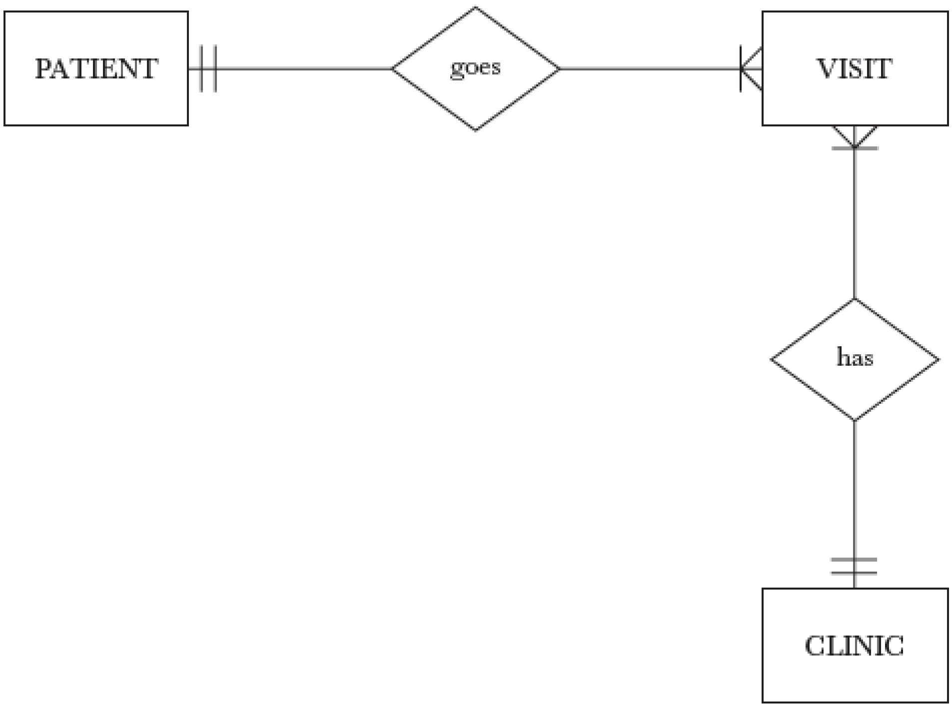


Figure 2: Entity Relationship Diagram

Entities The rectangles in the ERD represent entities. An entity is a person, place, or thing about which the organization wishes to store data. The entities depicted in the final version of the ERD will be transformed into tables in the relational database. Figure 8.4 shows an example of a table structure that might be created from the entity CLINIC. (Please note that these examples are quite simplistic and meant to illustrate general concepts rather than represent actual database design practices.)

Attributes: The attributes of an ERD can be shown as oval shapes extending from the entities; however, it is more common to see the entities listed separately or within the entity rectangle (see Figure 3). Attributes transform to data fields. Each entity in the ERD must have a unique identifier, called its *primary key*. The primary key cannot be duplicated within a table and cannot contain a null value. The primary key is also used to link entities together in order to form relationships.

PATIENT ENTITY ATTRIBUTES

PATIENT_MRN (PK)
PATIENT_LNAME
PATIENT_FNAME
PATIENT_MNAME
PATIENT_SSN
PATIENT_STREET
PATIENT_CITY
PATIENT_STATE
PATIENT_ZIPCODE
PATIENT_PHONE
PATIENT_DOB
...

VISIT ENTITY ATTRIBUTES

VISIT_ID (PK)
VISIT_DATE
VISIT_REASON
PATIENT_MRN (FK)
CLINIC_ID(FK)
...

CLINIC ENTITY ATTRIBUTES

CLINIC_ID (PK)
CLINIC_NAME
CLINIC_STREET
CLINIC_CITY
CLINIC_STATE
CLINIC_ZIPCODE
CLINIC_PHONE
CLINIC_FAX
...

Figure 3: Partial Attribute Lists for Patient, Clinic, and Visit

Relationships: Relationships within ERDs may be shown as diamond shapes. The name of the relationship is usually a verb. There are three possible relationships among any two entities: one-to-one, one-to-many, or many-to-many. Many-to-many relationships must be converted to one-to-many before a relational database can be implemented. In our ERD example (Figure 2), the one side of a relationship is shown by a single mark across the line and the many side is shown by a three-pronged crow's foot. To decipher the relationship between PATIENT and VISIT as shown in Figure 2, you would say, ''for each instance of PATIENT there are many possible instances of VISIT, and for each instance of VISIT there is only one possible instance of PATIENT.''

Preparing a data model to include only those relationships that can and should be implemented in the resulting database is called *normalization*. Normalizing the database ensures that data are stored in only one location in the database (except for planned redundancy). Storing each piece of data in only one location decreases the possibility of data anomalies as a result of additions and deletions. This reduction in data redundancy and decreased potential for data anomalies is the hallmark of a relational database. It is what distinguishes it from the flat file, an older database model.

Object-Oriented Databases

A newer database structure is the *object-oriented database* (OODB). The basic component in the OODB is an object rather than a table. An object includes both data and the relationships among the data in a single conceptual structure. An *object-oriented database management system* (OODBMS) uses *classes* and *subclasses* that inherit characteristics from one another in a hierarchical manner. Think, for example, of mammals as one class of animals in the physical world (with reptiles being another class) and humans as one subclass of mammals. Because all mammals have hair, humans "inherit" this characteristic. Object subclasses "inherit" properties from an object class in a similar manner. If a "person" object is defined as having a last name and a first name variable, then any subclass objects, such as "patient," will "inherit" these definitions.

The "patient" object may also have additional characteristics. A pure OODB is not common in the health care market, but products are beginning to incorporate elements of OODB and object-oriented programming with relational databases (Lee, 2002). The *object-relational database management system* (ORDBMS) is a product that has relational database capabilities plus the ability to add and use objects. One example on the market today is ObjectStore. The advantage of an ORDBMS is that many of the newer health care applications use video and graphical data, which an ORDBMS can handle better than a traditional RDBMS can. An ORDBMS also has the capability of incorporating hypermedia and spatial data technology. Hypermedia technology allows data to be connected in *web* formations, with hyperlinks. Spatial data technology allows data to be stored and accessed according to locations

Data Dictionaries

One very important step in developing a database to use in a health care application is the development of the *data dictionary*. The data dictionary gives both users and developers a clear understanding of the data elements contained in the database. Confusion about data definitions can lead to poor-quality data and even to poor decisions based on data misconceptions. A typical data dictionary allows for the documentation of

- Table names
- All attribute or field names
- A description or definition of each data element
- The data type of the field (text, number, date, and so forth)
- The format of each data element (such as DD-MM-YYYY for the date)
- The size of each field (such as 14 characters for a Social Security Number)
- An appropriate range of values for the field (such as integers 000000 to 999999 for a medical record number)
- Whether or not the field is required (is it a primary key or a linking key?)
- Relationships among fields

The importance of a well thought out data dictionary cannot be overstated. When an organization is trying to link or combine databases, the data dictionary is a vital tool.

Think, for example, how difficult it might be to combine information from databases with different definitions for fields with the same name.

Clinical Data Repositories

Many health care organizations, particularly those moving toward electronic medical records, develop *clinical data repositories*. Although these databases can take different forms, in general the clinical data repository is a large database that gets data from various data stores within application systems across the organization. There is generally a process by which data are *cleaned* before they are moved from the source systems into the repository. Once the clean data are in the repository, they can be used to produce reports that integrate data from two or more data stores.

Data Warehouses and Data Marts

A *data warehouse* is a type of large database designed to support decision making in an organization. Traditionally, health care organizations have collected data in a variety of *online transactional processing* (OLTP) systems, such as the traditional relational database and clinical data repository. OTLP systems are well suited for supporting the daily operations of a health care organization but less well suited for decision support. Data stored in a typical OLTP system are always changing, making it difficult to track trends over time, for example. The data warehouse, in contrast, is specifically designed for decision support.

Like a clinical data repository, a data warehouse stores data from other database sources. Creating a data warehouse involves extracting and cleaning data from a variety of organizational databases. However, the underlying structure of a data warehouse is different from the table structure of a relational database. This different structure allows data to be extracted along such dimensions as time (by week, month, or year), location, or diagnosis. Data in a data warehouse can often be accessed via *drill-down* menus that allow you to see smaller and smaller units within the same dimension. For example, you could view the number of patients with a particular diagnosis for a year, then a month in that year, then a day in that month. Or you could see how many times a procedure was performed at all locations in the health system, and then see the total by region, then by facility. Even though the same data might be available in a relational database, its normalized structure makes the queries you would have to use to get at the information quite complex and difficult to execute. Data warehouses help organizations transform large quantities of data from separate transactional files into a single decision-support database. ***Data marts*** are structurally similar to data warehouses but generally not as large. The typical data mart is developed for a particular purpose or unit within an organization.

Data Mining

Data mining is another concept closely associated with large databases such as clinical data repositories and data warehouses. However, data mining (like several other IT concepts) means different things to different people. Health care application vendors may use the term data mining when referring to the user interface of the data warehouse or data repository. They may refer to the ability to drill down into data as data mining, for example. However, more precisely used, data mining refers to a sophisticated analysis tool that automatically discovers patterns among data in a data store. Data mining is an advanced form of decision support. Unlike passive query tools, the data mining analysis tool does not require the user to pose individual specific questions to the database. Instead, this tool is programmed to look for and extract patterns, trends, and rules. True data mining is currently used in the business community for marketing and predictive analysis. This analytical data mining is, however, not currently widespread in the health care community.

NETWORKS AND DATA COMMUNICATIONS

The term *data communications* refers to the transmission of electronic data within or among computers and other related devices. Devices that make up computer networks must be compatible. They must be able to communicate with one another.

Network Communication Protocols

Data communication across computer networks is possible today because of *communication protocols and standards*. Without the common *language* of protocols, networked computers and other devices would not be able to connect with and talk to one another.

Network Types and Configurations

Computer networks used in health care and elsewhere are described with a variety of terms. a computer network is a collection of devices (sometimes called *nodes*) that are connected to one another for the purpose of transmitting data. A *network operating system* (NOS) is a special type of system software that controls the devices on a network and allows the devices to communicate with one another. Some of the most common network operating systems on the market today are Microsoft's Windows and Novell's NetWare

LAN Versus WAN The first distinction that is often made when describing a network is to identify it as either a *local area network* (LAN) or a *wide area network* (WAN.) LANs typically operate within a building or sometimes across several buildings belonging to a single organization and located in the same general vicinity. The actual distance a LAN covers can vary greatly. One common way to distinguish a LAN from a WAN is that the LAN will have its network hardware and software under the control of a single organization. As the Internet and its related technologies are used more by organizations, the line between LANs and WANs may be becoming somewhat blurred.

Topology A second way that wired networks are described is by their *topology*, or layout. There are two types of network topology: *physical* and *logical*. The physical topology is how the wires are physically configured. The logical topology is the way data flow from node to node in the network. Various arrangements and standards dictate this movement.

Network Media and Bandwidth

Two frequently discussed aspects of a network are its media and its bandwidth. *Media* refers to the physical "wires" or other transmission devices used on the network. *Bandwidth* is a measure of media capacity.

Media Data may be transmitted on a network through several types of media. Common types of conducted media for LANs include *twisted pair wire, coaxial cable*, and *fiber-optic cable*. Common wireless media include *terrestrial* and *satellite microwave transmissions* as well as *spread spectrum radio transmissions*. Mobile phone technology and infrared technology are also being used for wireless computer data transmission.

Service Carriers Communications across a WAN may involve some type of telecommunications carrier. These carriers provide the telephone lines, satellites, modems, and other services that allow data to be transmitted across distances. They can be either *common carriers*, primarily the long-distance telephone companies, or *special-purpose carriers*. Common carriers can provide either a traditional switched line, sometimes referred to as *plain old telephone service* (POTS), or a *dedicated* or *leased line*, which offers a permanent connection between two locations. Telephone companies also offer *integrated services digital network* (ISDN) services. ISDN uses existing phone lines to transmit not only voice but also video and image data in digital form. A purchased *T-1 line* may be another option for transmitting integrated voice, data, and images for large health care organizations, depending on their needs.

Bandwidth is another name for the capacity of a transmission medium. Generally, the greater the capacity, or bandwidth, of the medium, the greater the speed of transmission. Multiple factors influence transmission speed, and bandwidth is only one of them, but a low bandwidth can impede transmission rates across the network. Transmission rates are expressed as *bits per second* (bps). In other words, a medium's capacity is determined by the maximum number of bps it can carry.

Network Communication Devices

If you think about how computers are used in the health care organization today, they rarely depend on a single LAN to access all the information needed. At the least a computer will be connected to one LAN and the Internet. Often a single computer in a health care organization will be connected to multiple LANs and several WANs, including the Internet. LANs employ combinations of software and hardware in order to communicate with other networks.

There are several types of devices that allow networks to communicate with another, including *hubs, bridges, routers, gateways*, and *switches*.

INFORMATION PROCESSING DISTRIBUTION SCHEMES

Networks and databases are often described in terms of the method through which the organization distributes their information processing. Three common distribution methods are *terminal-to-host, file server*, and *client/server*. All three types are found in health care information networks. A single health care organization may in fact employ one, two, or all three methods of processing distribution, depending on its computing needs and its strategic decisions regarding architecture.

In ***terminal-to-host*** **schemes** the application and database reside on a host computer, and the user interacts with the computer using a dumb terminal, which is a workstation with no processing power. In some terminal-to-host setups the user may interact with the host computer from his or her personal computer (which obviously has computing power), but special software, called *terminal emulation* software, is used to make the PC act as if it were a dumb terminal when connecting to a specified host computer.

Thin client **schemes** are a variation of the host-to-terminal type. The major advantage cited for using this type of distribution is the centralized control. The individuals who support the network and databases no longer have to worry about PC maintenance or how the user might inadvertently modify the configuration of the workstation. *File server* systems have the application and database on a single computer. However, the end-user's workstation runs the database management system. When the user needs the data that reside on the file server, the file server will send the entire file to the computer requesting it.

***Client/server* systems** differ from traditional file server systems in that they have multiple servers, each of which is dedicated to one or more specialized functions. For example, servers may be dedicated to database management, printing, or other program execution. The servers are accessible from other computers in the network, either all computers in the network or a designated subset. The client side of the network usually runs the applications and sends requests from the applications to the server side, which returns the requested data.

CLINICAL AND MANAGERIAL DECISION SUPPORT

Health care executives and providers are faced with decisions every day, multiple times per day. The success of any health care organization literally depends on these large and small decisions. In this section we will describe technologies that support decision making in health care today, for both clinical and managerial decisions. The types of systems that we examine are

- Decision-support systems (DSS)
- Artificial intelligence systems, including expert systems, natural language processing, fuzzy logic, and neural networks

Nobel Prize–winning economist Herbert Simon described decision making as a three-step process. The steps involve

1. Intelligence: collecting facts, beliefs, and ideas. In health care these facts may be stored as data elements in a variety of data stores.

2. Design: designing the methods with which to consider the data collected during intelligence. These methods may be models, formulas, algorithms, or other analytical tools. Methods are selected that will reduce the number of viable alternatives.

3. Choice: making the most promising choice from the limited set of alternatives. Problems that face health care executives and clinicians may be *structured*, *unstructured*, or *semi-structured*. Structured problems are also referred to as *programmable* problems, because a computer program can be written with relative ease to solve this kind of problem. Transaction-based applications can be used to solve structured, or programmable, problems. For example, a payroll system is based on known facts about each employee's salary, deductions, and so on. The "decision" of how much to write the monthly paycheck for is fairly

straightforward. The unstructured and semi-structured problems present much more of a challenge for computer application developers.

Decision-Support Systems

How do we harness the power of a computer to solve a problem or make a decision about a solution when the situation is not easily structured with a simple algorithm (sequence of logical steps)? The computer systems developed to tackle the unstructured or semi-structured problem are called *decision-support systems* (DSS). Decision-support system is another term that can mean slightly different things to different vendors or users. In this section we are referring primarily to the traditional, stand-alone DSS: in other words, an application that is designed for the purpose of supporting decisions.

This is not the only form of decision support available to health care executives and providers today. For example, patient care or administrative applications may have components, such as data mining, that aid in decision making, but these applications might not be classified as full-blown DSS. An electronic spreadsheet, such as Excel, can also be used as a decision-support tool. Spreadsheets have built-in functions as well as the ability to use what-if statements.

Artificial Intelligence

Artificial intelligence (AI) is a branch of computer science devoted to emulating the human mind. One very common use of AI today is incorporated into the Google search engine. When the user types a misspelled word in a string of keywords, Google will suggest alternative keywords based on the context of the query. AI is a broad field with many different types of technology. Most AI is quite complex and describing the underlying technology is beyond the scope of this text.

TRENDS IN USER INTERACTIONS WITH SYSTEMS

This section of this chapter is devoted to describing some of the new and not-so-new devices that enhance the user interface with the health care information system. There have been many developments in input and output devices, along with personal computing devices, in the past few years. These developments are likely to continue and will affect the way in which users expect to interact with health information systems. The list of devices discussed in this section is by no means all inclusive. Each coming year will likely see new or improved devices on the market. However, these discussions will give you an overview of the various types of devices that are available at the time of this writing. We examine four categories of devices:

- Input devices
- Output devices

- External storage devices
- Mobile personal computing devices

Input Devices

The most common computer input devices in use today are the standard **keyboard** and the **mouse**. Other commonly used input devices and methods include **trackballs**, **trackpads**, **touch screens**, source data input devices such as **bar-code scanners**, and systems for imaging and speech recognition.

A special class of input devices known as *source data input* devices includes optical mark recognition, optical character recognition, and bar-coding devices, among others. Although bar coding has been commonplace in retail venues for many years, it has recently received a lot of attention in the health care community as a means of improving patient safety. *Optical bar-code recognition* devices recognize data encoded as a series of thick and thin *bars*. As with other technologies, the success of bar coding in health care stems from the development of standards, in this case the Health Industry Bar Code (HIBC) standard.

Many health care organizations have looked to *document imaging* systems as a means of getting data into health care information systems. Document imaging systems scan documents and convert them to digital images. These images are then stored in databases for later retrieval.

Speech recognition, or *voice recognition*, is another input method used in health care. It is particularly suited for situations or work environments where using a keyboard, mouse, or touch screen is not practical, such as the pathology lab or surgical suite.

Output Devices

The most commonly used computer output devices are the computer monitor and the printer. Notebook and handheld computing devices rely on flat screen technology, and they have also become a popular alternative to the CRT for desktop computers.

Speech output is another form of computer output that is becoming more commonplace. Automated telephone answering systems employ computer speech output, for example. There are two approaches to speech output: in one, phrases prerecorded by a person are strung together to form the output desired; in the other, *synthesized speech*, a machine produces the speech sounds.

External Storage Devices

Health care information systems require the extensive use of external storage devices. Critical systems must be backed up regularly, and data must be frequently archived for permanent or nearly permanent storage. Among the newer types of external storage is optical disk technology, which is available in such forms as *compact disks* (CDs), *digital video disks* (DVDs), and *optical tape*.

Flash memory is another form of external storage (and internal storage in many handheld computer devices) that is gaining popularity as its costs come down. Compared to other external storage, flash memory can be accessed more rapidly, consumes less power, and is smaller in size. The disadvantage is its comparative cost .

Mobile Personal Computing Devices

Many types of mobile personal computing and handheld devices are used in health care today. In fact many health care organizations have had to respond to providers who have adopted *personal digital assistants* (PDAs) and *pocket PCs* and who subsequently want and expect to be able to access health care applications from these devices.

A ***laptop*** **(or** ***notebook***) *computer* is a compact, lightweight personal computer. The screen and keyboard are built in. Although they still lag somewhat behind desktop computers in speed, memory, and capacity, today's laptops are powerful enough to replace traditional desktop computers. PDAs allow users not only to store data, such as calendars and personal notes, but also to connect to the Internet to browse Web pages and send or receive e-mail. These devices are becoming more powerful and less expensive, which will likely increase their popularity. More software applications are being developed specifically for PDAs. In the health care community, resources such as medical dictionaries, formularies, and clinical coding systems can be installed on PDAs. As PDAs and cellular phones both gained popularity, the market recognized the potential for incorporating aspects of both into a single device, sometimes called a *smart phone*. These devices are evolving rapidly and are being employed in health care organizations.

Chapter 9

Electronic Health Records

Objectives

- Learn the basics of EHR functionality
- Learn the benefits, both clinical and financial, of using an EHR
- Become familiar with basic EHR terms
- Learn the characteristics of a successful implementation
- Learn about the must-have features of an EHR

Overview of the EHR

The Electronic Health Record (EHR)– then called the Electronic Medical Record (EMR) or Computerized Patient Record (CPR)– received its first real validation in an Institute of Medicine's (IOM) report in 1991 entitled "The Computer-Based Patient Record: An Essential Technology for Health Care.(www.nap.edu)" IOM drove home the idea that the EHR is needed to transform the health system to improve quality and enhance safety.

The EHR is about quality, safety, and efficiency. It is a great tool for physicians, but cannot ensure these virtues in isolation. Achieving the true benefits of EHR systems requires the transformation of practices, based on quality improvement methodologies, system and team based care, and evidence-based medicine.

Basic Terminology

The following is a list of basic terms you will need to know as you navigate the EHR market:

- **Certificatio**n - This relates to a national effort to "certify" various requirements for EHR software. The Certification Committee for Health Information Technology (CCHIT) is tasked with determining what basic "must have" features EHR systems contain in order to be "certified."

- **Electronic Health Record (EHR) -** This term refers to computer software that physicians use to track all aspects of patient care. Typically this broader term also encompasses the practice management functions of billing, scheduling, etc.

- **Electronic Medical Record (EMR)** - This is an older term that is still widely used. It has typically come to mean the actual clinical functions of the software such as drug interaction checking, allergy checking, encounter documentation, and more.

- **Integrated EHR** - This refers to an EHR that is integrated with practice management software. Typical choices include purchasing a fully integrated product which performs all the functions of practice management software, or a stand-alone EHR which is compatible with an existing practice management system.

- **Structured and unstructured data entry** - There are several ways of entering data into your EHR as you practice. These include dictating straight into the software (voice recognition), templates, and writing (handwriting recognition).

- **Templates** - Pre-structured portions of the software for common and/or basic visits. These templates fill in a standard set of data which you may then customize for each individual visit. Templates can be used with dictation, writing, or choosing among a menu of options formulated for each specific template.

Potential Benefits of an EHR

Benefits of an EHR can be categorized as follows:

1. Potential Productivity and Financial Improvement

- Fewer chart pulls
- Improved efficiency of handling telephone messages and medication refills
- Improved billing
- Reduced transcription costs
- Increased formulary compliance and clearer prescriptions leading to fewer pharmacy call backs
- Improved coding of visits

Additional potential benefits may include: population management and proactive patient reminders; improved reimbursement from payers due to EHR usage; and participation in pay-for-performance programs.

2. Quality of Care Improvement

- Easier preventive care leading to increased preventive care services
- Point-of-care decision support
- Rapid and remote access to patient information
- Easier chronic disease management
- Integration of evidence-based clinical guidelines
-

3. Job Satisfaction Improvement

- Fewer repetitive, tedious tasks
- Less "chart chasing"
- Improved intra-office communication
- Access to patient information while on-call or at the hospital
- Easier compliance with regulations
- Demonstrable high-quality care

4. Customer Satisfaction Improvement

- Quick access to their records
- Reduced turn-around time for telephone messages and medication refills
- A more efficient office leads to improved care access for patients
- Improved continuity of care (fewer visits without the chart)
- Improved delivery of patient education materials

Things to Think About Before EHR Implementation

The difference between an efficient, harmonious, computerized office and computerized chaos is all about implementation. It is useful to think that your office will not just be computerized, but you will be implementing a complete health information technology system. If you think in terms of the new system and how everything works within the system, success is much easier.

No paper charts the prime directive, and all good things flow from this accomplishment. Certainly there are other accomplishments such as efficient workflow, elimination of wasted steps, and improved office communication– but none of this can happen if paper charts are not eliminated. We continue to live in a paper-based world and many strategies will be needed to deal with the great amount of paper coming into and flowing from your office.

Office/Physician Characteristics of a Successful EHR Implementation

Many characteristics enable successful implementations of information technology projects (or any office projects, for that matter) are human rather than technological factors.

Medical offices that have had successful EHR implementations have the following characteristics:

1. Excellent teamwork
2. Excellent communication skills
3. A spirit of adventure and continual improvement
4. The ability to handle adversity and bumps in the road well.
5. Excellent problem solving skills
6. A willingness and flexibility for individuals to go outside their job descriptions in order to make things work.

Family physicians who have had successful electronic health record implementations have the following characteristics:

1. An interest in how their entire practice works as well as parts of the practice that are inefficient and may be subject for improvement with an electronic health record system.
2. A teamwork mentality and the ability to trust employees as well as delegate appropriately.
3. An appreciation of the work involved in transitioning from paper charts to electronic charts and a willingness to perform this work.
4. A willingness to change their work process from a paper-based system to a new work process that takes advantage of information technology.

Things to Think About Before Going Live with Your EHR

1. Workstation Setup

Will you be using workstations in the examination rooms or will you be using wireless laptops or tablets that you carry around?
If you'll be using workstations, pay strict attention to workstation set up in the room so that you can interact with the computer and interact with the patient in a reasonable way. One good idea that seems to work well for many offices is putting the flat screens on a swivel arm so that they can be readjusted easily.

2. Network Setup

Accommodations need to be made in the room for ethernet connectors, outlets, etc. Do not be too concerned about patients and children messing with the computers. This has been found by many to be a minimal problem. If you are using a tablet or laptop computer you need to be concerned about your wireless set up and make sure this works appropriately.

3. Power/Operation Issues

Battery life is also a concern and many have docking stations within the room to provide a stable platform for entering data. Other offices use mobile carts to carry the tablets or laptops from room to room. Your ancillary staff will also need mobile computers. This may seem obvious, but test your hardware extensively and make sure it works reliably and quickly before going live. Also test your electronic health record software extensively and make sure it works well with the hardware that you have selected.

4. EHR/Practice Management Integration

If at all possible, you will want to have your practice management and billing system connected to your electronic health record. Many electronic health records are installed as a complete solution containing both practice management and electronic medical record. If this is not the case, an interface could possibly be built between the two pieces of software.

5. Lab Connectivity

Establish digital connections with the laboratories you deal with, if possible. These may also require an interface. Interfaces can be expensive, but the flow of information in your office and the digital format of data that come from it (especially laboratory) makes it very worthwhile. Having digital connections from end to end and information flowing into your office in a digital format at the time that you start to implement your record makes things much easier.

6. Dealing With Paper from Outside Sources

You may be on the verge of having a paperless office, but most of the world is still using paper. Therefore, you must think extensively about how you will deal with the paper that presently comes into your office and the paper that you produce that goes out of your office.

There are multiple ways to deal with this, some more efficient than others.

- **Faxing** - The ability to fax prescriptions and other electronic documents is an absolute essential element to your strategy.

- **Scanning -** You also need a robust scanning solution as paper will be coming into your office from the mail which will need to be scanned into the record.

- **Messaging -** Your electronic health record should contain a robust messaging system for interoffice messaging. After all, you no longer have paper charts to attach sticky notes. Robust messaging capability is absolutely essential and is the glue that holds everything together especially within a larger office. You will find it essential.

- **Printing -** While you want to minimize the amount of paper coming into your office either through the front door or from the paper fax machine, you still need the capability of printing prescriptions, patient information, and other documents. Consider carefully where you want to position your printers within the office space. Many practices have found it useful to have printers in each examination room where prescriptions, instructions, and information can be printed real-time and handed directly to the patient.

Electronic Health Record Features & Functions

Basic EHR Functions:

Some of these functions include, but are not limited to:

- Identify and maintain a patient record
- Manage patient demographics
- Manage problem lists
- Manage medication lists
- Manage patient history
- Manage clinical documents and notes
- Capture external clinical documents
- Present care plans, guidelines, and protocols
- Manage guidelines, protocols and patient-specific care plans
- Generate and record patient-specific instructions

Select the Right Electronic Health Record for Your Practice

Developing an EHR Evaluation Matrix

An evaluation matrix is a simple grid. Across the top are the criteria you want in your EHR and down the side are the EHRs you're considering. You can enter the information about how each EHR meets your criteria and easily compare products. You can also send the empty matrix to each vendor you are considering and ask them complete it.

When EHR certification standards become a reality in the not-too-distant future, it will be easier to evaluate products based on their functionality. Then the evaluation matrix becomes easier and criteria such as vendor stability, compatibility, and interoperability will move to the forefront of the evaluation process.

Understanding EHR Demonstrations

Several options are available to see an EHR demonstrated. Each method has its pros and cons, and you may choose to use more than one in your evaluation process.

Non-interactive video: This requires the fewest resources from the practice, but it is also offers the least benefit. This type of video can give you clues into the complexity of the user-interface and a glimpse into the functionality. Keep in mind that the video was produced by the vendor to showcase the best of the product and what the vendor thinks is important. This type of demonstration is useful when your EHR list of contenders is long.

Interactive trial demo: This requires more time compared with non-interactive video. Your goal during the demonstration should be to gauge the ease of use. The questions you want answered are:

- Is the product intuitive?
- Is it easy and quick to perform repetitive tasks (e.g., writing a prescription)?
- Is it easy to find specific patient information?

Vendor directed demo: This will require more time than an interactive trial demo and you will need to schedule it with the vendor. Your goal during the demonstration should be to understand the true functionality available in the system. (If you have not already had an interactive demo, you will need to answer those questions as well.) Come prepared with your list of criteria from your evaluation matrix. You want to see how the functions you require are accomplished. Also bring any issues you uncovered during the interactive trial demo.

Besides the functionality, you want to see the workflow. The easiest way to accomplish this is to bring some clinical scenarios that mirror what you see in practice (e.g., acute visit for URI; chronic disease care for new and follow-up patients; routine visits such as annual or well child exam). Study how these mock patients flow through the system and how the data is entered and viewed by front desk, nurse, physician, billing, etc.

Live site demo: This requires the most resources from the practice, but offers the largest reward in terms of getting data about an EHR. Because of the time, travel, and money involved in this type of demonstration, many physicians limit these to their final few contenders. To set up a live site demo, you first need to find peers like yourself that are using the EHR in their practice. You can ask the vendor to give you a list of family physicians using their EHR in your region. By asking a vendor for a list, however, you accept the likelihood that these users are happy with their EHR– since it is in the vendor's best interest. You can mitigate this by doing two things:

1. Ask for a list of ten to 20 users

2. Briefly interview the users by telephone before scheduling a site visit

Asking for such a large list and interviewing users can help you find some who may have had a problem with the EHR. You want to ask questions about how they deal with negatives of the system and how well they can explain the functions of the system. This demonstration should build on the previous demonstrations. If you have the resources, doing more than one site visit for a particular EHR is beneficial.

Chapter 10

Data privacy and Security

Objectives

- Know the Important of Security and privacy in HIS
- List the Pillars od security
- The important of staff rules to preserve Security

The Importance of HIS Security and Privacy

Patients are vulnerable in health care settings, not just because they're seeking medical attention, but also due to the fact that they're sharing private information with their health care organization. Helping patients feel comfortable enough to share this data allows professionals to provide the most efficient care and can strengthen the patient-physician bond. Protecting their information is also preserving their rights, individuality, respect and dignity, as stated in "Beyond the HIPAA Privacy Rule: Enhancing Privacy, Improving Health Through Research" by the Institute of Medicine.

But it goes beyond sharing information via word-of-mouth. Patients rely on their data to be stored and processed in a secure computer system, free of security breaches. As health IT becomes more complex and the distribution of information through electronic systems becomes more common, health care professionals are responsible for taking extra precautions to ensure privacy.

Patients must trust that their health information is secure and private to reap the benefits of digital health technology. If patients do not feel that their information is kept securely, they may not be willing to disclose information, which could keep them from receiving the care they need. When organizations don't have strong security and privacy practices in place, it puts your health information system at a heightened risk of experiencing a cyber-attack. This can jeopardize the reputation of the facility and put patients' health at risk.

There are three major ethical priorities for electronic health records: privacy and confidentiality, security, and data integrity and availability.

Privacy and Confidentiality

Justices Warren and Brandeis define privacy as the right "to be let alone". According to Richard Rognehaugh, it is "the right of individuals to keep information about themselves from being disclosed to others; the claim of individuals to be let alone, from surveillance or interference from other individuals, organizations or the government". The information that is shared as a result of a clinical relationship is considered confidential and must be protected. The information can take various forms (including identification data, diagnoses, treatment and progress notes, and laboratory results) and can be stored in multiple media (e.g., paper, video, electronic files). Information from which the identity of the patient cannot be ascertained—for example, the number of patients with prostate cancer in a given hospital—is not in this category.

Patient information should be released to others only with the patient's permission or as allowed by law. This is not, however, to say that physicians cannot gain access to patient information. Information can be released for treatment, payment, or administrative purposes without a patient's authorization. The patient, too, has federal, state, and legal rights to view, obtain a copy of, and amend information in his or her health record.

The key to preserving confidentiality is making sure that only authorized individuals have access to information. The process of controlling access—limiting who can see what—begins with authorizing users. In a physician practice, for example, the practice administrator identifies the users, determines what level of information is needed, and assigns usernames and passwords. Basic standards for passwords include requiring that they be changed at set intervals, setting a minimum number of characters, and prohibiting the reuse of passwords. Many organizations and physician practices take a two-tier approach to authentication, adding a biometrics identifier scan, such as palm, finger, retina, or face recognition.

Security

The National Institute of Standards and Technology (NIST), the federal agency responsible for developing information security guidelines, defines information security as the preservation of data confidentiality, integrity, availability (commonly referred to as the "CIA" triad).

The increasing concern over the security of health information stems from the rise of EHRs, increased use of mobile devices such as the smartphone, medical identity theft, and the widely anticipated exchange of data between and among organizations, clinicians, federal agencies, and patients. If patients' trust is undermined, they may not be forthright with the physician. For the patient to trust the clinician, records in the office must be protected. Medical staff must be aware of the security measures needed to protect their patient data and the data within their practices.

A recent survey found that 73 percent of physicians text other physicians about work. How to keep the information in these exchanges secure is a major concern. There is no way to control what information is being transmitted, the level of detail, whether communications are being intercepted by others, what images are being shared, or whether the mobile device is encrypted or secure.

Integrity and Availability

In addition to the importance of privacy, confidentiality, and security, the EHR system must address the integrity and availability of information.

Integrity assures that the data is accurate and has not been changed. This is a broad term for an important concept in the electronic environment because data exchange between systems is becoming common in the health care industry. Data may be collected and used in many systems throughout an organization and across the continuum of care in ambulatory practices, hospitals, rehabilitation centers, and so forth. This data can be manipulated intentionally or unintentionally as it moves between and among systems.

Poor data integrity can also result from documentation errors, or poor documentation integrity. A simple example of poor documentation integrity occurs when a pulse of 74 is unintentionally recorded as 47. Whereas there is virtually no way to identify this error in a manual system, the electronic health record has tools in place to alert the clinician that an abnormal result was entered.

Availability. If the system is hacked or becomes overloaded with requests, the information may become unusable. To ensure availability, electronic health record systems often have redundant components, known as fault-tolerance systems, so if one component fails or is experiencing problems the system will switch to a backup component.

Improve patient data security

Practices often lack basic security policies and procedures, allow staff members to share passwords, and fail to turn on or properly configure the security features of their electronic health record (EHR) systems. In addition, many practices fail to perform security risk assessments, despite a requirement to do so under the Health Insurance Portability and Accountability Act (HIPAA).

Here are 10 steps that experts say can help practices defend their protected health information (PHI) and their businesses from cyber criminals.

1- Do a security risk assessment

Security risk assessments must be performed annually. If a consultant is required, a security risk assessment can cost several thousand dollars. There are also costs for security risk mitigation. For example, practices might have to buy extra software to supplement their EHR's security tools, which may cover only some aspects of security.

2- Encrypt data

Patient data should be encrypted whenever possible. Any current certified EHR can perform this task. Experts agree that while encryption is essential, practices should not rely on this approach alone—or on other technical fixes such as antivirus programs and firewalls—to defend the privacy and security of data. The weak point of encryption is that it relies on protecting access to the system. If a password is stolen, for example, the thief can use that password to access data, whether or not it is encrypted. More than 80% of security breaches result from human factors. While few practice staffers would steal PHI, they could unwittingly introduce malware into a practical network by falling fir phishing emails or tactics.

3- Control system access

Access control, a key component of security, takes different forms depending on a practice's network and how its EHR and practice management system are hosted.

In a client-server network, where the server that stores the EHR is located on-site, the providers and staff access the EHR through their computer network. If a practice uses a cloud-based EHR, in contrast, the application and the data are stored on a remote server, and individual workstations and other computers reach the EHR through a web browser.

Cloud-based EHR vendors configure the security features for their servers, "so it's more seamless and you don't have to worry about that [part of security]". It can be tricky for practices to configure those features properly in client-server networks, they may have to hire outside help

On the other hand, the cloud vendor's security protects only the remote server, not the practice IT infrastructure, she points out. If somebody steals a password, or if malware gets into the network, the practice's data security is still at risk.

Other deficiencies in basic security can also leave data vulnerable. Many practices, for example, don't apply security patches to their computer operating systems. Moreover, many groups are still using outmoded operating systems such as Windows XP, which Microsoft no longer supports and represents an invitation to hackers. No practice concerned about security should stick with these older operating systems.

4- Authenticate users

Most EHRs authenticate users with a login name and a password. Experts say practices should frequently change passwords and make them complex enough to foil hackers.

Change passwords every 60 to 90 days. Even then, hackers using a “brute force” attack—in which the attacking computer runs through a large number of combinations—may be able to figure out a password.

A better approach is two-factor authentication. This method couples a password with biometric identification, such as a thumbprint, a text that a user has to respond to or some other factor that only authorized users can provide.

5- Provide remote access securely

Providers may need remote access from home or other locations to do their work. If a practice has a cloud-based EHR, users with remote privileges can access the EHR in the usual way through their web browser.

In a practice with a client-server network, however, remote users must access the network to get to the EHR. If a user’s home computer is infected with malware, a robust firewall coupled with antivirus and intrusion detection software will help keep it out of the system. But cyber thieves can still steal data during remote user sessions if the link with the network is insecure.

To prevent this, experts recommend using a virtual private network (VPN) that encrypts all of the data in transit and disappears after a session is over. A VPN is a secure, temporary computer-to-computer connection within the public internet. Practices can easily download VPN software, but need someone with technical expertise to install and configure it.

6- Adopt role-based access

Most EHRs allow practices to configure their software to limit different levels of the system to employees who need to use that portion of the application and view the associated data.

For example, in an EHR integrated with a practice management system, a receptionist may only need to use the scheduling application; role-based access would not let that person access any clinical or financial data.

This approach helps protect privacy and prevent the use of PHI to commit fraud. In addition, if a user’s password is stolen and that person has only partial access to the EHR, it limits how much damage the thief can do.

7-Don’t store data on user devices

“What small providers have in place in terms of security isn’t typically there to keep information confidential, but to protect their access to it.” That could explain why many small practices allow users (staff, physicians and anyone with access to the system) to store PHI on their desktop computers, laptops and even mobile devices. But doing so makes the information more vulnerable to hackers.

“If the employees aren’t aware of the importance of PHI being centralized, and they start creating PDFs and performing print screens that means that information is still being stored locally,” he points out. Staff should be instructed to access information from the network and not store any data on their own devices.

Many doctors and nurses use tablets and smartphones at work. These devices can pose security threats if they’re allowed to connect to the network or if they can download EHR data.

Some healthcare organizations install remote “wiping” software on laptops and mobile devices employees use at work. In theory, that would allow a practice to erase data on those devices if they were lost or stolen.

Use and scan audit logs

All certified EHRs have audit logs that record which user did what in the EHR and when. However, practices often don’t turn these logs on or configure them correctly. According to Kim, a recent HIMSS survey showed that fewer than half of hospitals and practices were using their audit logs or another feature designed to prevent people from tampering with the logs to erase the signs of an intruder. Practices that don’t know how to activate and configure audit logs should ask their vendors or IT consultants about it. Beyond that, she notes, practices need software that automatically scans their audit logs to detect anomalies that might indicate a cyberattack, such as an unfamiliar user or a known user logging on at an unusual time of the day.

Back up data off site

Practices that use a client-server system should have onsite backup, such as a mirrored server that can replace the main server if it goes down. In addition, all practices should have off site backup, both for security purposes and in case of natural disasters. In addition, practices should maintain off site copies of their financial data, including data from billing systems, general ledgers and payroll systems.

A cloud-based EHR vendor or hosting firm will back up EHR data, says Nussbaum. Practices that have client-server systems should back up their data on a tape and move it offsite at least daily. It’s essential to keep these backups offline in case a hacker takes over your network. Also, backups should be encrypted. Otherwise, a lost backup tape is considered a security breach under HIPAA.

Get business associate agreements

HIPAA requires practices to sign business associate agreements (BAA) with all outside parties with which they share PHI.

These agreements obligate the business associates to safeguard the PHI. Organizations covered by HIPAA do not have to evaluate the security procedures of their business associates, but some experts suggest that practices question business associates about their security practices in general to help safeguard data.

Hashey does this before he signs a BAA, mainly to ensure that outside firms understand the importance of protecting patient information.

Sacopulos agrees this is a good idea, but cautions against including business associates in security risk assessments. It's impractical because it involves too many entities. Also, if a practice signs off on a business associate's security practices, it's assuming a legal duty that it's not obligated to take on.

Your responsibility as a health care professional

Providers and health care professionals alike, such as a specialist in health informatics, have a responsibility to comply with privacy and security requirements. Whether you're dealing with electronic health records, customer bills, or other secure information, it is your job to ensure it has limited access and is kept safe under standard administrative processes. Health informatics professionals can help to mitigate these risks and ensure data is secured

Chapter 11

E-governance and Management

Objectives

- To be able to understand the scope and importance of information technology governance.
- To review the IT roles and responsibilities of users, the IT department, and senior management.
- To review the factors that enable sustained excellence in the application of IT.

Overview

In this chapter we discuss an eclectic but important set of information technology (IT) governance. Developing, managing, and evolving IT governance and management mechanisms is often a central topic for organizational leadership.

E-GOVERNANCE

Electronic governance or e-governance is the application of information and communication technology (ICT) for delivering government services, exchange of information, communication transactions, integration of various stand-alone systems and services between government-to-citizen (G2C), government-to-business (G2B), government-to-government (G2G), government-to-employees (G2E) as well as back office processes and interactions within the entire government framework. Through e-governance, government services are made available to citizens in a convenient, efficient, and transparent manner. The three main target groups that can be distinguished in governance concepts are government, citizens, and businesses/interest groups. In e-governance there are no distinct boundaries.

Health Governance

The World Bank's 2004 World Development Report, focusing on accountability structures and processes, highlights the link between governance and sectoral service delivery. Since then, there has been a growing recognition that governance interventions can contribute to service delivery improvements and that technical elements need to be accompanied by governance elements to strengthen public services. USAID's HFG project supports country governments to both integrate governance in their health sector reform activities and improve specific aspects of health governance.

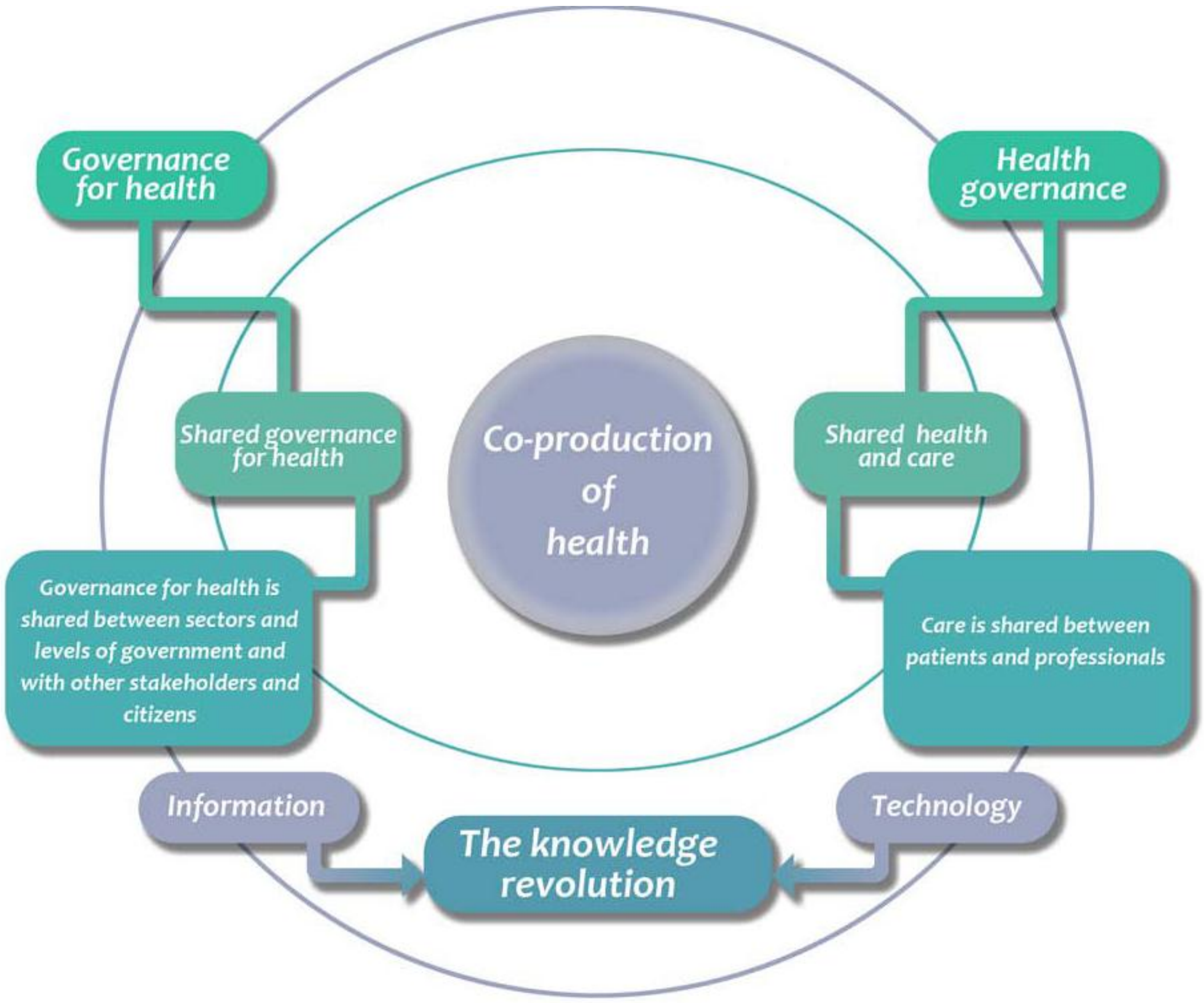


Figure 1: Governance for health and health governance

Governance for Health

governance for health" as the attempts of governments or other actors to steer communities, countries or groups of countries in the pursuit of health as integral to wellbeing through both a 'whole-of-government' and a 'whole-of-society' approach.

- It positions health and well-being as key features of what constitutes a successful society and a vibrant economy in the 21st century and grounds policies and approaches in values such as human rights and equity.
- Governance for health promotes joint action of health and non-health sectors, of public and private actors and of citizens for a common interest. It requires a synergistic set of policies, many of which reside in sectors other than health as well as sectors outside of government, which must be supported by structures and

mechanisms that enable collaboration.

- It gives strong legitimacy to health ministers and ministries and to public health agencies, to help them reach out and perform new roles in shaping policies to promote health and wellbeing.

Good governance for Health

Several international organizations have drawn up principles of good governance for governments, including the EU, the OECD and the World Bank. To some extent, these principles have emerged in parallel to guidelines for good corporate governance as standards for the behavior of companies. More recently, such standards are also being applied in nongovernmental organizations. Understanding of good governance in relation to governments is well captured in the World Bank's definition of governance (World Bank, 2011)

Figure 2: Good Environment *Source*: UN DP (1997).

Smart Governance for Health

In a knowledge society, policy decisions based purely on normative considerations lose ground to decisions based on evidence. At the same time, decision-making requires new methods for coping with and accounting for the uncertainties that abound when knowledge – always questionable, always revisable – supersedes majority values as the basis for authority. Smart governance is one way of describing the major institutional adaptations being undertaken in public and international organizations in the face of increasing interdependence. Smart governance, a term coined by Willke (2007), is "an abbreviation for the ensemble of principles, factors and capacities that constitute a form of governance able to cope with the conditions and exigencies of the knowledge society" (Figure 3)

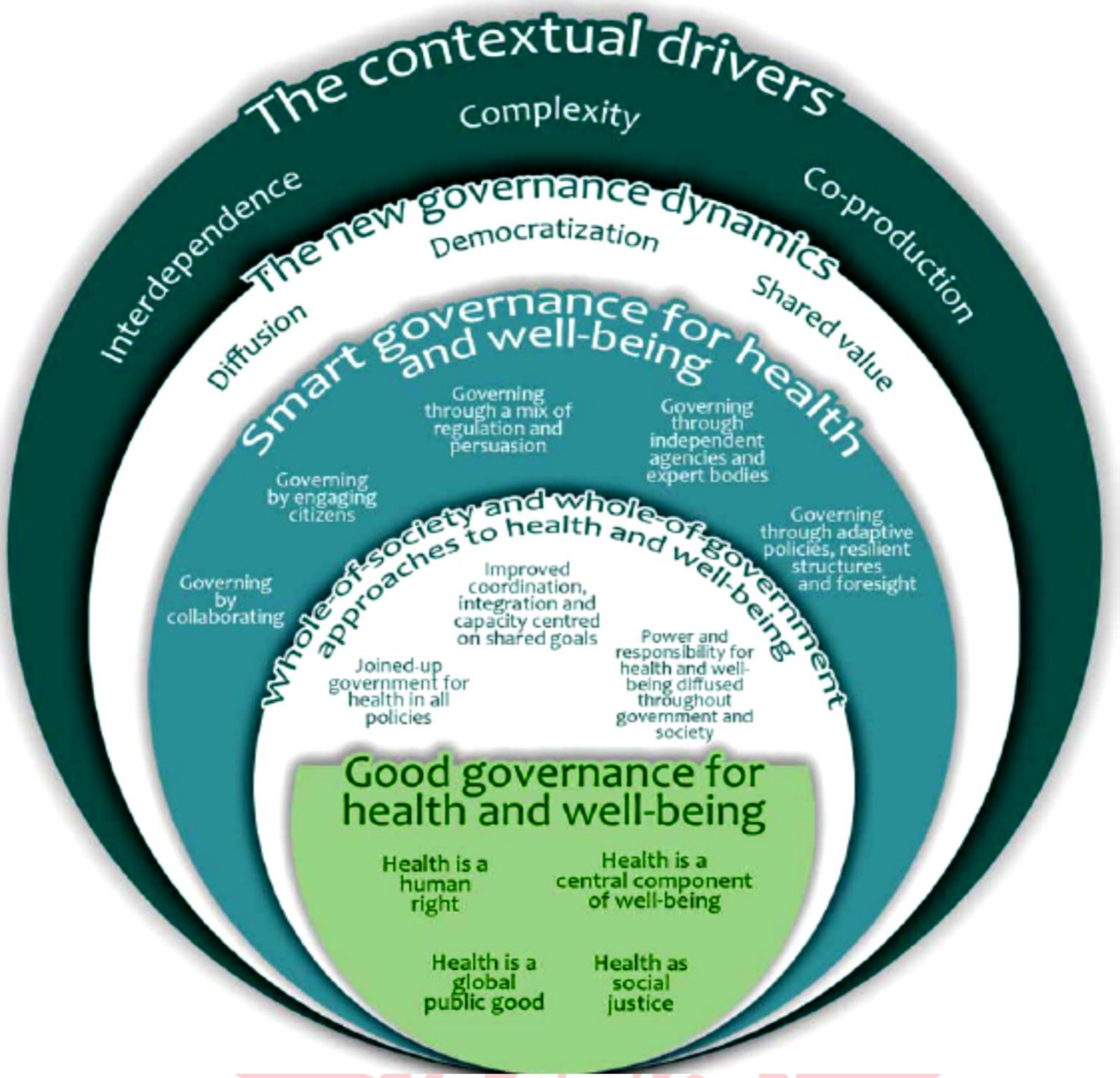


Figure. 3. Smart governance for health

Five types of smart governance for health and well-being

Smart governance for health defines how governments approach governance for health challenges strategically in five dimensions, through:

- collaboration;
- engagement;
- a mixture of regulation and persuasion;
- independent agencies and expert bodies; and
- adaptive policies, resilient structures and foresight.

FOUNDATION OF IT GOVERNANCE

There are five major areas that form the foundation of IT governance. The organization's governance mechanisms need to create structures and processes for these areas.

- IT principles: high-level statements about how IT is used in the business.
- IT architecture: an integrated set of technical choices to guide the organization in satisfying business needs. The architecture is a set of policies, procedures, and rules for the use of IT and for evolving IT in a direction that improves IT support for the organization.
- IT infrastructure strategies: strategies for the existing technical infrastructure (and IT support staff) that ensure the delivery of reliable, secure, and efficient services.
- Business application needs: processes for identifying the needed applications.
- IT investment and prioritization: mechanisms for making decisions about project approvals and budgets.

Developing and maintaining an effective and efficient IT governance structure is a complex exercise. Moreover, governance is never static. Continuous refinements may be needed as the organization discovers imperfections in roles, responsibilities, and processes.

Governance Characteristics

Well-developed governance mechanisms have several characteristics. *They are perceived as objective and fair*. No organizational decision-making mechanisms are free from politics, and some decisions will be made as part of "side deals." It is exceptionally rare for all managers of an organization to agree with any particular decision. No matter how good an individual is at performing his or her IT governance role, there will be members of the organization who will view that individual as a lower life form. Nonetheless, organizational participants should generally view governance as fair, objective, well-reasoned, and having integrity. The ability of governance to govern is highly dependent on the willingness of organizational participants to be governed.

IT, User, and Senior Management Responsibilities

Effective application of IT involves the thoughtful distribution of IT responsibilities between the IT department, users of applications and IT services, and senior management. In general these responsibilities address decision-making rights and roles. Although different organizations will arrive at different distributions of these responsibilities, and an organization's distribution may change over time, there is a fairly normative distribution.

IT Department Responsibilities:

The IT department should be responsible for the following:

- Developing and managing the long-term architectural plan and ensuring that IT projects conform to that plan.

- Developing a process to establish, maintain and evolve IT standards in several areas: Telecommunications protocols and platforms

    - Client devices, e.g., workstations and PDAs, and client software configurations

    - Server technologies, middleware and database management systems

    - Programming languages

    - IT documentation procedures, formats and revision policies

    - Data definitions (this responsibility is generally shared with the organization function, e.g., finance and health information management, that manages the integrity and meaning of the data)

    - IT disaster and recovery plans

    - IT security policies and incident response procedures

- Developing procedures that enable the assessment of sourcing options for new initiatives, e.g., build vs. buy new applications or leveraging existing vendor partner offerings versus utilizing a new vendor when making an application purchase

- Maintaining an inventory of installed and planned systems and services and developing plans for the maintenance of systems or the planned obsolescence of applications and platforms

- Managing the professional growth and development of the IT staff
- Establishing communication mechanisms that help the organization understand the IT agenda, challenges and services and new opportunities to apply IT
- Maintaining effective relationships with preferred IT suppliers of products and services.

User Responsibilities

IT users (primarily middle managers and supervisors) have several IT-related responsibilities:

- Understanding the scope and quality of IT activities that are supporting their area or function
- Ensuring that the goals of IT initiatives reflect an accurate assessment of the function's needs and challenges and that the estimates of the function's resources (personnel time, funds and management attention) needed by IT initiatives, e.g., to support the implementation of a new system, are realistic
- Developing and reviewing specifications for IT projects and ensuring that ongoing feedback is provided to the IT organization on implementation issues, application enhancements and IT support, e.g., ensuring that the new application has the functionality needed by the user department
- Ensuring that the applications used by a department are functioning properly, e.g., by periodically testing the accuracy of system-generated reports and checking that passwords are deleted when staff leave the organization
- Participating in developing and maintaining the IT agenda and priorities. These responsibilities constitute a minimal set. In Chapter Seven, we discussed an additional, and more significant, set of responsibilities during the implementation of new applications.

Senior Management Responsibilities

The primary IT responsibilities of the senior leadership are as follows:

- Ensuring that the organization has a comprehensive, thoughtful and flexible IT strategy
- Ensuring an appropriate balance between the perspectives and agendas of the IT organization and the users, e.g., the IT organization may want a new application

that has the most advanced technology while the user department wants the application that has been used in the industry for a long time

- Establishing standard processes for budgeting, acquiring, implementing and supporting IT applications and infrastructure

- Ensuring that IT purchases and supplier relationships conform to organizational policies and practices, e.g., contracts with IT vendors need to use standard organizational contract language

- Developing, modifying and enforcing the responsibilities and roles of the IT organization and users

- Ensuring that the IT applications and activities conform to all relevant regulations and required management controls and risk mitigation processes and procedures

- Encouraging the thoughtful review of new IT opportunities and appropriate IT experimentation [Applegate, Austin, & McFarlan, 2003, p. 68].

Chapter 12

Applications of HISs

Objectives

- Be familiar with common applications in healthcare organization
- Learn the alternative application and its features
- Learn the benefits of Medical applications

Overview

There are seemingly an endless number of possible applications of information technology (IT) to health service management. Enthusiasm in introducing IT solutions in health care is sometimes bypassing traditional scrutiny and quality control. Without proper assessment and system thinking (how implementation, in part of the system, may sometimes produce negative effects in other parts of the system) one should not introduce new IT solutions.

The area of e-health is as said, very broad, covers topics such as telemedicine, electronic records, recruitment, going paperless, procurement, healthcare score cards, audits, information systems etc. There are three areas of health informatics:

- Consumer informatics
- Medical and clinical informatics, and
- Bio-informatics.

These categories are based on the predominant type of user or use.

Consumer informatics

Often this category – *Consumer Informatics* – is the one commonly referred to as 'e-health' and focuses communications to patients and the public about health topics.

Consumer-to-consumer (C-to-C) applications are potentially strong means of empowering individuals and the public. There are 25,000 – 30,000 health-oriented websites and they are among the most visited. These sites are and will be major sources of information and misinformation. There is an urgent need for all concerned, including politicians/lawmakers, health professionals and industry to put in place adequate standards and quality control for these websites.

Medical / clinical informatics

This category relates directly to health care structure, processes and outcomes. A main application is **computer-based medical records**, a sub-category of which is *computer-based personal records* that will facilitate access to low cost therapies, for example, with certain areas of mental health, such as depression.

Another sub-category is *computer-based patient records* that will facilitate clinical decision-making. These later records may be linked to knowledge-oriented systems that may contribute to quality control of clinical processes. Such a decision support has been demonstrated to have improved outcomes.

Computer-based population or community health records are usually anonymized patient and/or personal records. These systems are particularly valuable in public health where one is trying to trace different types of health hazards, linked either to medical, environmental or social agents.

What general comments, therefore, can be made regarding computer-based records? There is a certainly important ethical concern in relation to composition of records and access to the same. Also, linking different record systems to each other sometimes raises criticism, in particular in cases, which may involve personal/patient records. Again, there is need to secure standards and qualities and for appropriate steps, nationally and internationally, to be taken in the search for solutions. Also, lack of guidance from central authorities, have in many instances led to a mish-mash of non-compatible computer-based patient record systems. Such circumstances have caused problems to arise in the smooth processing of patients between health service units, even within the same health authority (or equivalent).

Telemedicine

Finally, telemedicine provides a category by itself. Telemedicine, meaning healthcare delivered by electronic means, has been on the road for over a century – if care provided by telegraph and telephone is considered. However, towards the end of the last century, this emerged as a delivery system with huge potential due to the information technology revolution, which made two-way, audio-visual transmission possible at reasonable cost. It has a long way to go before it can be effectively integrated into a healthcare delivery system. One crucial difficulty is that many telemedicine applications have yet to be developed, evaluated and implemented in the hospital environment, before application of the system over longer distances.

Common Features of Medical Software

Electronic medical records	Electronic medical record (EMR) or electronic health record (EHR) software assists in creating and storing digital patient records. Helps track patient notes, demographics, histories and medications. Features include e-prescribing, SOAP notes, E&M coding advice and more. EMRs may also provide medical lab integration, device integration, tablet support and voice recognition.
Medical billing	Manages the creation of patient statements and submission of claims. Functions include coding, claim scrubbing, eligibility inquiry, electronic claim submission, payment posting and reporting.
Patient scheduling	Automates the process of scheduling patient visits. Features include automated follow-ups, text message/phone/email reminders and multi-location support. Typically offered with billing in a practice management suite.
Radiology information systems	Manages the operations and workflow of radiology imaging centers. Automates the process of storing, manipulating and distributing patient data and images.
Picture archiving and communications systems	Manages the storage and retrieval of DICOM images (X-rays, CAT scans, MRIs etc.). Often used in conjunction with an RIS to execute the radiology workflow efficiently.
Medical accounting	Automates accounting procedures for healthcare practices. Functions include A/R, A/P, general ledger, financial reporting and more.
Clinic management	Combines practice management software and EMR software to handle the business and practitioner sides of a clinic.
Prescription writing	Helps doctors and practices create, print, record and transmit prescriptions by offering a group of dedicated applications and software add-ons.
Patient engagement	Allows doctors to stay in communication with their patients by providing educational resources and improving patient-provider relationships.
Practice analytics	Tracks data for doctors and practices such as patient intake, revenue cycle, reimbursement rates, and other information to help give an understanding of overall operations.

What are the Consumer requirements

Most organizations are researching and evaluating medical software for one or more of the following reasons:

Transitioning from paper charts to digital records. “It’s raining paper” is the common cry we hear from paper-based practices. These buyers want to cut back on paper, improve office efficiency, reduce errors and run a more effective operation overall.

Replacing outdated software. This is a common scenario we hear from buyers. Their current system—whether it be a homegrown system or from a medical software vendor—is out of date and costly to maintain or update. They want a more modern system that is easier to use, meets federal requirements (e.g., ONC-ATCB certification) or that meets feature/functional needs.

Combining applications into an integrated suite. In many cases these practices have a hodgepodge of disparate applications, and as a result, find themselves doing double data entry and dealing with other inefficient workflows and processes. These organizations invest in integrated medical office management software—that is, integrated EMR, billing and scheduling applications—to centralize all information and functions in one place.

Implementing best-of-breed applications. Conversely, these buyers are focused on applications to address a specific need. Most often, buyers in this category are looking for a stand-alone billing, EMR, RIS or PACS system.

Pursuing federal incentives. Thanks to the HITECH Act of 2009, physicians have been replacing their EHRs or purchasing new ones for the first time to meet federal requirements. In order to qualify for Medicare and Medicaid incentives, physicians—or more accurately, “eligible professionals”—must make “meaningful use” of a certified EHR. The law offered incentives for physicians who complied before 2015, but physicians who still aren’t meeting “meaningful use” standards today face penalties in the form of decreased reimbursements.

We should note that outpatient and inpatient organizations often have different feature/functional requirements. For example, inpatient care provider centers such as hospitals will require systems to support bed management, UB-04 billing and potentially long-term patient stays. Meanwhile, ambulatory care providers such as primary care physicians and specialists will share common feature requirements to support “walk-in/walk-out” care.

Practices looking to integrate business intelligence tools into their existing medical solutions might be interested in healthcare BI software.

Benefits of Medical Software

The general benefits of any medical system are improved quality of patient care, increased operational efficiency and improved practice profitability. These benefits are created by different applications and impact organizations in different ways. For example:

- The automation of back-office operations streamlines administrative tasks associated with patient encounters, which may enable providers to spend more time with patients and hire fewer staff.

- More accurate documentation of these encounters and a more organized claims submission process can lead to improved collections.

- Automated alerts prompt providers with potential issues or risks, while automated reminders help patients return to the office when necessary, improving quality of care.

Important Considerations

Integrated suite vs. best-of-breed. When selecting a system, buyers will have the choice of implementing different applications for specific tasks, or a complete suite of tools to address all their needs. The key decision that most providers will need to make is whether to implement a standalone electronic medical records (EMR) system or replace an existing practice management system with a complete system. We hear from many buyers facing this decision as practice management systems have been ubiquitous since the 1990s and EMRs are increasing in adoption, primarily due to the HITECH Act.

Software-as-a-Service (SaaS). The trend toward cloud computing is impacting many industries, and healthcare is certainly one of them. Web-based, or SaaS, software offers several advantages such as lower upfront costs, reduced IT and support costs, remote accessibility and more. However, practices in rural settings may not have access to the broadband Internet necessary to efficiently run Web-based software. Moreover, Web-based systems may not support all the feature/functional needs of some practices with unique requirements.

Mobile EHR Software. Going hand-in-hand with SaaS, healthcare providers are finding themselves increasingly on the go and accessing systems from multiple offices, home and mobile devices. Tablet (e.g., iPad) and smartphone support, including iPhones and Android phones, is increasingly common. If you will be accessing your software primarily from a mobile device, we suggest choosing a vendor that has developed a native app for your device, such as MediTouch's iPad EMR.

ONC-ATCB certification. As most healthcare professionals are aware, the HITECH Act of 2009 requires the use of electronic medical records systems by 2015. Eligible professionals can subsequently qualify for up to $44,000 through the Medicare EHR Incentive Program or up to $63,750 through the Medicaid EHR Incentive Program. To qualify, they will need to demonstrate “meaningful use” of one of the ONC-ATCB certified EMRs.

Chapter

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